Exact Second-Order Asymptotics for the Gaussian Wyner–Ziv Problem

Jinpeng Xu and Lin Zhou

Abstract

We revisit the Wyner–Ziv problem of lossy compression of Gaussian sources under quadratic distortion and characterize its exact second-order asymptotics for fixed distortion strictly between zero and the conditional source variance. Specifically, under a prescribed excess-distortion probability, the optimal second-order coding rate is determined by a joint Gaussian probability optimized over an allocation chosen from the observed source energy. For achievability, we restrict a Gaussian auxiliary distribution to a source-dependent shell and apply the Poisson matching bound, followed by a conditional Gaussian approximation. For the converse, we control the posterior distribution of an arbitrary measurable code and derive two likelihood bounds in which the additional observation and description costs cancel. The achievability and converse analyses yield the same optimized Gaussian probability, averaged over the source-energy fluctuation. Furthermore, we show that side information available only at the decoder incurs a strictly positive second-order rate loss relative to side information at both terminals, although the first-order rates coincide. An explicit example illustrates this loss.

Index Terms

Lossy source coding, rate-distortion, Wyner–Ziv coding, Gaussian sources, side information, second-order asymptotics.

I. Introduction

Lossy source coding, also known as rate-distortion theory, was introduced by Shannon [1] to characterize the compression rate required when a prescribed reconstruction distortion is permitted. The rate-distortion function describes the first-order asymptotics as the blocklength tends to infinity. To approximate the performance of optimal codes at finite blocklengths, Ingber and Kochman [2] and Kostina and Verdú [3, Theorem 12] derived second-order asymptotics under an excess-distortion probability criterion. The second-order coding rate specifies the leading correction to the first-order rate for a fixed, nonvanishing excess-distortion probability.

Beyond point-to-point compression, Wyner and Ziv [4, Theorem 1] studied lossy source coding with side information available only at the decoder. The encoder compresses a source sequence into a finite message, and the decoder reconstructs the source using the message and a correlated observation. For jointly Gaussian sources under quadratic distortion, the Wyner–Ziv rate-distortion function equals the conditional rate-distortion function with side information at both terminals [4, Sec. I-C, Remark 6], [5]. Le, Tan, and Motani [6, Theorem 2] subsequently characterized the second-order coding rate for the both-terminal setting. The equality of the first-order rates leaves open whether withholding the side information from the encoder incurs a second-order rate loss.

For the discrete Wyner–Ziv problem, Zhou [7, Corollary 1] characterized the exact second-order asymptotics under a channel-overlap condition in the interior distortion regime. The characterization allows the encoder to choose an allocation between information and distortion constraints according to the observed source. However, the finite-alphabet arguments do not directly apply to Gaussian sources with unbounded quadratic distortion. A Gaussian characterization requires a source-dependent construction and an arbitrary-code converse that retain the joint information and distortion fluctuations while respecting the encoder's observation restriction.

A. Main Contributions

We address the above question and derive the exact second-order coding rate for fixed positive source and noise variances, distortion strictly between zero and the conditional source variance, and excess-distortion probability in $(0, 1)$. For achievability, we first define a Gaussian auxiliary distribution and restrict the difference between the source and auxiliary description to a shell whose radius depends on the observed source energy. We then prove a uniform lower bound on the shell probability and calculate the resulting conditional information and distortion means. The conditional Poisson matching bound and a joint Gaussian approximation yield the achievable second-order rate.

For the converse, we begin with an arbitrary measurable code, select source sequences on which its conditional success probability is sufficiently large, and control the posterior covariance by adding a small Gaussian observation. We derive upper and lower likelihood bounds that retain the exact costs of the added observation and sampled source description. After these costs cancel, the same Gaussian optimization as in achievability bounds the success probability. Finally, we prove that the resulting second-order coding rate strictly exceeds the both-terminal benchmark and cannot, for all error probabilities, be expressed as the square root of a scalar dispersion multiplied by a standard Gaussian quantile. We also derive a representation using the absolute value of a Gaussian variable for a symmetric family and an explicit example illustrating the rate loss.

B. Other Related Works

Nonasymptotic and second-order achievability bounds for coding with side information were derived by Watanabe, Kuzuoka, and Tan [8]. Li and Anantharam [9, Lemma 2 and Theorem 3] developed a Poisson matching approach that permits distinct encoder and decoder reference distributions. We use this coding bound after specifying the source-dependent Gaussian auxiliary. Li and Li [10] refined finite-alphabet achievability through type deviation convergence. The Gaussian shell estimates in the present paper account for the continuous source alphabet, while the converse retains the restriction that the encoder cannot observe the side-information noise.

C. Organization for the Rest of the Paper

In Section II, we formulate the Gaussian Wyner–Ziv problem, recall the first-order result, and define the second-order coding rate. In Section III, we present and discuss our main result, including the comparison with both-terminal side information and a numerical example. The achievability and converse proofs are provided in Sections IV and V, respectively. Finally, Section VI concludes the paper and discusses future directions. The proofs of supporting results are deferred to the appendices.

Notation

Random variables and their realizations use uppercase and lowercase letters, respectively. We use $\mathbb{N}$ and $\mathbb{R}$ for the positive integers and real numbers, $[n] := \{1, \dots, n\}$, and $X^n := (X_1, \dots, X_n)$ for $n \in \mathbb{N}$. Calligraphic letters denote sets. All logarithms and $\exp$ use base e. The standard Gaussian CDF and density are denoted by Φ and φ, respectively; Φ^{-1} denotes the inverse CDF. We use $\mathcal{N}(m, V)$ for a Gaussian distribution with mean m and covariance V, and 0_n and I_n for the zero vector and identity matrix of order n, respectively.

The symbols $\langle \cdot, \cdot \rangle$, $\| \cdot \|$, and $\| \cdot \|_{\mathrm{F}}$ denote the Euclidean inner product, Euclidean norm, and Frobenius norm, respectively. We write $\mathbb{1}\{\cdot\}$ for an indicator and $\mathbb{P}$, E, Var, and Cov for probability, expectation, variance, and covariance, and $\mathrm{KL}(P\|Q)$ for relative entropy. We use H, h, and I for entropy, differential entropy, and mutual information, respectively. We use Cov for a covariance matrix. The governing distribution is specified by a subscript when needed. A subscript on a distribution denotes its marginal or conditional distribution; for example, P_{XY} denotes the joint distribution of (X, Y), and $P_{XY}^{\otimes n}$ denotes its n-fold product. The symbols p_X and $p_{X|Y}$ denote densities when they exist. We write $C^\top$, $\mathrm{tr}(C)$, and $\det C$ for the transpose, trace, and determinant of a matrix C, respectively; $C \succeq 0$ means that C is positive semidefinite.

The symbols $\Rightarrow$ and $\xrightarrow{P}$ denote convergence in distribution and probability. We use $\sigma(\cdot)$ for the generated sigma-field. The symbols $O(\cdot)$ and $o(\cdot)$ have their usual deterministic asymptotic meanings; subscripts specify permitted dependence of their constants. For positive a_n, the notation $W_n = o_P(a_n)$ means that W_n/a_n converges to zero in P-probability, while $W_n = O_P(a_n)$ means that this ratio is bounded in P-probability. The same convention applies under other governing distributions. Proof-specific notation is defined locally.

II. Problem Formulation and Existing Results

A. System Model and Definitions

Fix source and noise variances $p, q > 0$ and a blocklength $n \in \mathbb{N}$. Under the original distribution P, let $(X_i, Z_i)_{i \in [n]}$ be independent copies of independent variables $X \sim \mathcal{N}(0, p)$ and $Z \sim \mathcal{N}(0, q)$, and let

$$Y^n = X^n + Z^n. \tag{1}$$

For one coordinate, write $Y := X + Z$. The source, side-information, and reproduction alphabets are $\mathbb{R}$. For any $(x^n, \hat{x}^n) \in \mathbb{R}^n \times \mathbb{R}^n$, define

$$d(x^n, \hat{x}^n) := \frac{1}{n} \sum_{i \in [n]} (x_i - \hat{x}_i)^2 \tag{2}$$

$$= \frac{1}{n} \|x^n - \hat{x}^n\|^2, \tag{3}$$

where (3) is the definition of the Euclidean norm. As shown in Fig. 1, only the decoder observes Y^n. The compressed message is received without transmission error.

Definition 1. *For $(n, M_n) \in \mathbb{N}^2$, an (n, M_n)-code consists of a measurable encoder*

$$f : \mathbb{R}^n \to [M_n] \tag{4}$$

and a measurable decoder

$$\phi : [M_n] \times \mathbb{R}^n \to \mathbb{R}^n. \tag{5}$$

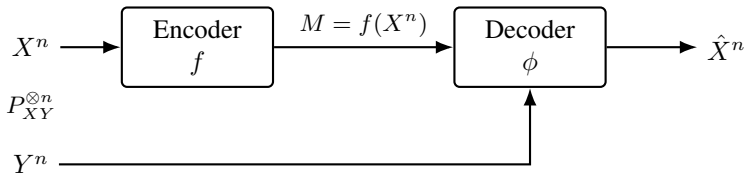


Fig. 1. System model for quadratic Gaussian Wyner–Ziv coding. The source and side information satisfy (1); the side information is available only at the decoder.

The transmitted message is $M = f(X^n)$ and the reconstruction is $\hat{X}^n = \phi(M, Y^n)$. Given $D > 0$, the excess-distortion probability is

$$P_{\mathrm{e}}^{(n)}(D) := P\{d(X^n, \hat{X}^n) > D\}. \tag{6}$$

For $(D, \varepsilon) \in (0, \infty) \times (0, 1)$, define the finite-blocklength fundamental limits

$$M_n^\star(D, \varepsilon) := \min\{M_n \in \mathbb{N} : \text{an } (n, M_n)\text{-code satisfies } P_{\mathrm{e}}^{(n)}(D) \le \varepsilon\}. \tag{7}$$

The corresponding minimum rate, in nats per source symbol, is

$$R_n^\star(D, \varepsilon) := \frac{1}{n} \log M_n^\star(D, \varepsilon). \tag{8}$$

We also compare with the setting in which both terminals observe Y^n. In that setting, the encoder may use (X^n, Y^n), while the decoder and excess-distortion criterion remain as in Definition 1.

B. Existing First-Order Asymptotics

Define the regression coefficient

$$\rho := \frac{p}{p+q}. \tag{9}$$

For the jointly Gaussian pair in (1), its conditional mean is

$$\mathsf{E}[X \mid Y] = \rho Y, \tag{10}$$

where (10) follows from Gaussian regression with covariance p and observation variance $p + q$. Define the conditional source variance by

$$S := \mathsf{Var}(X \mid Y) \tag{11}$$

$$= p - \frac{p^2}{p+q} \tag{12}$$

$$= \frac{pq}{p+q}, \tag{13}$$

where (12) follows from Gaussian regression with source variance and source-observation covariance both equal to p, and observation variance $p + q$; (13) simplifies the fraction. Throughout the paper, p, q, the distortion D, and the excess-distortion probability ε are fixed independently of n, with $0 < D < S$ and $\varepsilon \in (0, 1)$. This is the positive-rate interior. The additive model represents every nondegenerate correlated scalar jointly Gaussian pair after centering and an invertible rescaling of the side information.

Let $R_{\mathrm{WZ}}(D)$ denote the Wyner–Ziv rate-distortion function. The Gaussian Wyner–Ziv formula [4, Sec. I-C, Remark 6], [5] and the conditional Gaussian converse [6, Theorem 2 and Sec. VIII-C2] yield, for each fixed $\varepsilon \in (0, 1)$,

$$\lim_{n\to\infty} R_n^\star(D, \varepsilon) = R_{\mathrm{WZ}}(D) \tag{14}$$

$$= \frac{1}{2} \log \frac{S}{D}, \tag{15}$$

where the upper bound in (14) follows from Gaussian Wyner–Ziv achievability, the lower bound follows by supplying Y^n to the encoder and applying the cited conditional converse, and (15) is the Gaussian rate-distortion formula. The limit is independent of ε, which gives the strong converse. Both-terminal side information has the same first-order rate.

C. Definitions of Second-Order Asymptotics

Using the first-order rate in (15), we define the second-order coding rate under the excess-distortion probability criterion.

Definition 2 (Second-Order Coding Rate)**.** *A real number L is said to be second-order (D,ε)-achievable if there exists a sequence of (n, M_n)-codes such that*

$$\limsup_{n\to\infty} \frac{\log M_n - nR_{\mathrm{WZ}}(D)}{\sqrt{n}} \le L \tag{16}$$

and

$$\limsup_{n\to\infty} P_{\mathrm{e}}^{(n)}(D) \le \varepsilon. \tag{17}$$

The infimum of all such numbers is denoted by $L^{\star}_{\mathrm{WZ}}(D,\varepsilon)$ and is called the optimal second-order coding rate. When the encoder also observes Y^n, denote the analogously defined optimal second-order coding rate by $L_{\mathrm{both}}(\varepsilon)$.

The both-terminal benchmark is [6, Theorem 2]

$$L_{\mathrm{both}}(\varepsilon) = \frac{1}{\sqrt{2}}\Phi^{-1}(1-\varepsilon). \tag{18}$$

We use (18) to quantify the second-order cost of restricting the side information to the decoder.

III. Second-Order Asymptotics

A. Preliminaries

Recall the regression coefficient ρ, conditional source variance S, and first-order rate $R_{\mathrm{WZ}}(D)$ in (9), (13), and (15), respectively. We first introduce the Gaussian random variables and the optimization that determine the second-order coding rate. Define the distortion multiplier

$$\lambda := -R'_{\mathrm{WZ}}(D) \tag{19}$$

$$= \frac{1}{2D}, \tag{20}$$

where (20) follows by differentiating (15). For $x^n \in \mathbb{R}^n$, define the source-energy fluctuation

$$A_n(x^n) := \frac{1-\rho}{2\sqrt{n}}\left(\frac{\|x^n\|^2}{p} - n\right). \tag{21}$$

Let $A \sim \mathcal{N}(0, (1-\rho)^2/2)$. Under the source distribution in (1),

$$A_n(X^n) \Rightarrow A, \tag{22}$$

where (22) follows from the central limit theorem since the independent variables X_i^2 have mean p and variance $2p^2$.

Let (J, B) be a centered Gaussian pair, independent of A, with covariance

$$\Sigma := \begin{pmatrix} \left(\rho - \frac{D}{q}\right)\left(1 - \frac{1}{2}\left(\rho - \frac{D}{q}\right)\right) & -\frac{D}{2q}\left(\rho - \frac{D}{q}\right) \\ -\frac{D}{2q}\left(\rho - \frac{D}{q}\right) & \frac{D}{q}\left(1 - \frac{D}{2q}\right) \end{pmatrix}. \tag{23}$$

The matrix Σ is positive definite for $0 < D < S$, as proved in Appendix A. The random variables J and B represent the information fluctuation and the distortion fluctuation multiplied by λ, respectively. These fluctuations arise from the side-information noise when the source sequence and its auxiliary description are fixed. Given any $(r, s) \in \mathbb{R}^2$, define their joint CDF as

$$\Phi_{\Sigma}(r, s) := \mathbb{P}\{J \le r, B \le s\}. \tag{24}$$

Given any $z \in \mathbb{R}$, define the largest joint probability when the two thresholds sum to z:

$$H^{\star}(z) := \sup_{s\in\mathbb{R}} \Phi_{\Sigma}(z - s, s). \tag{25}$$

Increasing s permits a larger distortion fluctuation and a smaller information fluctuation. For a second-order rate $L \in \mathbb{R}$, the source fluctuation A leaves the threshold sum $L - A$. Thus, define

$$F_G(L) := \mathsf{E}[H^{\star}(L - A)], \tag{26}$$

and define the corresponding second-order rate by

$$L_G(D,\varepsilon) := \inf\{L \in \mathbb{R} : F_G(L) \ge 1 - \varepsilon\}. \tag{27}$$

Dependence on the fixed variances (p, q) is suppressed.

Proposition 1. *Fix any $p, q > 0$ and $0 < D < S$. The following results hold.*

(i) *The supremum in* (25) *admits a Borel maximizing selection $s^\star(z)$ bounded on compact intervals.*

(ii) *The function F_G is continuous and strictly increasing from zero to one. Hence $L_G(D, \varepsilon)$ is the unique solution of $F_G(L) = 1 - \varepsilon$.*

The proof of Proposition 1 is provided in Appendix A. The first statement permits the encoder to choose the allocation from the observed source energy. The second statement ensures that the equation $F_G(L) = 1 - \varepsilon$ determines a unique second-order rate.

B. Main Results and Discussions

Recall the optimal second-order coding rate $L^\star_{\mathrm{WZ}}(D, \varepsilon)$ in Definition 2 and the quantile $L_G(D, \varepsilon)$ defined in (27). Our main result is stated as follows.

Theorem 1. *Fix any $p, q > 0$, $0 < D < S$, and $\varepsilon \in (0, 1)$. The optimal second-order coding rate for the quadratic Gaussian Wyner–Ziv problem is*

$$L^\star_{\mathrm{WZ}}(D, \varepsilon) = L_G(D, \varepsilon). \tag{28}$$

The achievability and converse proofs of Theorem 1 are provided in Sections IV and V, respectively. We provide a proof sketch here. In the achievability part, we restrict a Gaussian auxiliary distribution to a shell whose radius depends on the observed source energy. The radius determines the conditional information and distortion means, and the Poisson matching bound reduces the error analysis to their joint fluctuations. A conditional Gaussian approximation then gives the probability in (25). In the converse part, we start from an arbitrary code and modify the observation so that its posterior mean and covariance can be controlled. We derive two likelihood bounds in which the added observation and description costs cancel. The remaining fluctuations have the same Gaussian limit as in achievability. Averaging over the source energy gives F_G in (26) for both directions.

Theorem 1 gives the following implications. Firstly, the optimization in (25) is performed for each value of A before taking the expectation in (26). The encoder may choose s from $A_n(X^n)$, but cannot use the unobserved side-information noise. Taking a union of the events $\{J \le z - s, B \le s\}$ over s would instead allow the choice to depend on (J, B). The source-dependent choice in our result is analogous to the type-dependent construction in the discrete problem [7, Sec. V-A]; the present proofs establish the shell estimates and posterior bounds for Gaussian sources and quadratic distortion.

Secondly, the decoder-only and both-terminal problems have different second-order coding rates. Recall $L_{\mathrm{both}}(\varepsilon)$ in (18). The following corollary makes the comparison precise.

Corollary 1. *Under the hypotheses of Theorem 1, for all $L \in \mathbb{R}$ and $\varepsilon \in (0, 1)$,*

$$F_G(L) < \Phi(\sqrt{2}L), \tag{29}$$

and hence

$$L^\star_{\mathrm{WZ}}(D, \varepsilon) > L_{\mathrm{both}}(\varepsilon). \tag{30}$$

The proof of Corollary 1 is provided in Appendix B. For every z, the event $\{J \le z - s^\star(z), B \le s^\star(z)\}$ has strictly smaller probability than $\{J + B \le z\}$. Averaging this inequality over A gives (29), since $A + J + B$ is centered Gaussian with variance $1/2$.

Corollary 1 shows that withholding the side information from the encoder incurs a second-order rate loss, although the first-order rates are equal. In particular, $L^\star_{\mathrm{WZ}}(D, 1/2) > 0$. For every scalar constant $V_{\mathrm{s}} \ge 0$, however, $\sqrt{V_{\mathrm{s}}}\Phi^{-1}(1/2) = 0$. Consequently, the optimal second-order coding rate cannot have the form $\sqrt{V_{\mathrm{s}}}\Phi^{-1}(1 - \varepsilon)$ for all $\varepsilon \in (0, 1)$.

1) Numerical Example: Finally, we specialize Theorem 1 to an example with a closed-form second-order rate. Let $X \sim \mathcal{N}(0, 2)$ and $Z \sim \mathcal{N}(0, 2)$ be independent, let $Y = X + Z$, and choose $D = 1/2$. Then $S = 1$ and $\rho = 1/2$, so the first-order rate in both settings is

$$R_{\mathrm{WZ}}(1/2) = \frac{1}{2}\log 2. \tag{31}$$

For this model, Theorem 1 simplifies to

$$L^\star_{\mathrm{WZ}}(1/2, \varepsilon) = \Phi^{-1}(\sqrt{1 - \varepsilon}), \tag{32}$$

where the Gaussian calculation is provided in Appendix B. The both-terminal second-order coding rate is given by (18).

To illustrate the rate loss in Corollary 1, Fig. 2 plots the following second-order rate approximations against the blocklength for $\varepsilon = 0.1$:

$$R^{(2)}_{n,\mathrm{WZ}} := \frac{1}{2}\log 2 + \frac{\Phi^{-1}(\sqrt{1 - \varepsilon})}{\sqrt{n}} \tag{33}$$

and

$$R^{(2)}_{n,\mathrm{both}} := \frac{1}{2}\log 2 + \frac{\Phi^{-1}(1 - \varepsilon)}{\sqrt{2n}}. \tag{34}$$

The second-order coding rates are approximately 1.6322 and 0.9062 nats, respectively. Thus, the decoder-only approximation exceeds the both-terminal approximation by approximately $0.7260/\sqrt{n}$ nats per source symbol. The difference decreases with the blocklength, while both approximations converge to the same first-order rate. The curves describe second-order asymptotic approximations.

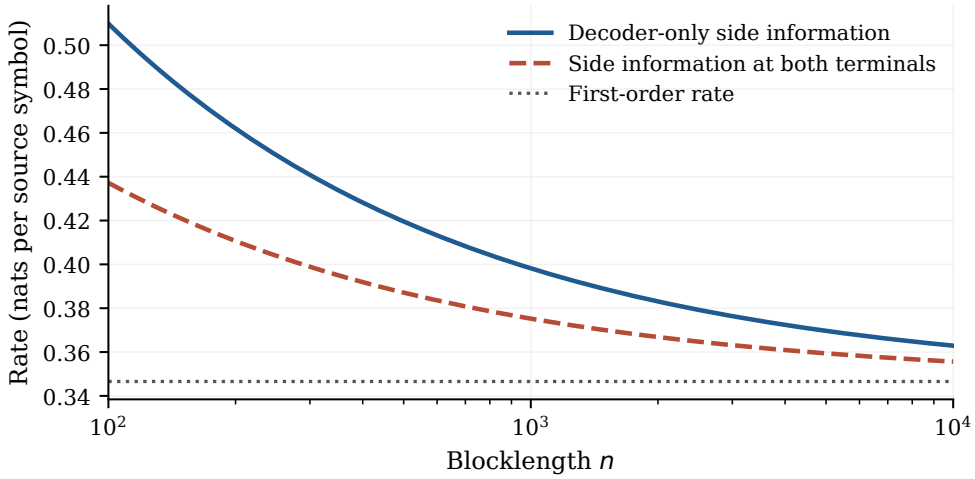


Fig. 2. Second-order rate approximations with decoder-only and both-terminal side information for $p = q = 2$, $D = 1/2$, and $\varepsilon = 0.1$. Both settings have first-order rate $\frac{1}{2}\log 2$; their second-order difference decreases as $n^{-1/2}$.

IV. Achievability Proof

Fix any $L \in \mathbb{R}$. We prove the achievability part of Theorem 1 by constructing deterministic codes satisfying

$$\log M_n \le nR_{\mathrm{WZ}}(D) + \sqrt{n}L + O(\log n). \tag{35}$$

Their excess-distortion probabilities satisfy

$$\limsup_{n\to\infty} P_{\mathrm{e}}^{(n)}(D) \le 1 - F_G(L). \tag{36}$$

We first construct a source-dependent Gaussian auxiliary and verify the hypotheses of the one-shot coding bound. We next evaluate the joint information and distortion constraints using a conditional Gaussian approximation. Finally, we average over the source energy and remove the source cutoff.

A. Preliminary Definitions

Define the residual variance

$$T := \frac{Dq}{q - D}. \tag{37}$$

The interior condition $0 < D < S$ implies $0 < T < p$. Under the Gaussian reference distribution $\bar{Q}$, let the scalar variables $U \sim \mathcal{N}(0, p - T)$ and $W_0 \sim \mathcal{N}(0, T)$ be independent, set $X = U + W_0$, and generate $Y = X + Z$ with $Z \sim \mathcal{N}(0, q)$ independent of (U, W_0). This reference has the prescribed source marginal and satisfies the Markov relation $U - X - Y$. Gaussian conditioning gives $\mathrm{Var}_{\bar{Q}}(X|U,Y) = D$ because the posterior precision is $T^{-1} + q^{-1} = D^{-1}$. The conditional mutual information is consequently $I_{\bar{Q}}(X;U|Y) = R_{\mathrm{WZ}}(D)$. For scalar (u, y), define the conditional-mean reconstruction

$$\psi(u, y) := \frac{D}{q}y + \left(1 - \frac{D}{q}\right)u. \tag{38}$$

The conditional distributions under $\bar{Q}$ are

$$\bar{Q}_{U|X=x} = \mathcal{N}\left(\left(1 - \frac{T}{p}\right)x, T\left(1 - \frac{T}{p}\right)\right), \tag{39}$$

$$\bar{Q}_{U|Y=y} = \mathcal{N}\left(\frac{p - T}{p + q}y, \frac{(p - T)(q + T)}{p + q}\right), \tag{40}$$

and

$$\bar{Q}_{X|U=u,Y=y} = \mathcal{N}(\psi(u, y), D), \tag{41}$$

where (39)–(41) follow from Gaussian conditioning in the preceding reference model and the definition of T in (37). Denote their densities by $\bar{q}_{U|X}$, $\bar{q}_{U|Y}$, and $\bar{q}_{X|U,Y}$, respectively, and write $\bar{q}_{X|Y}$ for the density of $\mathcal{N}(\rho y, S)$, which is also the original conditional distribution of X given $Y = y$. For $(x, u, y) \in \mathbb{R}^3$, define the information score g. Bayes' rule gives

$$g(x,u,y) := \log \frac{\bar{q}_{U|X}(u|x)}{\bar{q}_{U|Y}(u|y)} \tag{42}$$

$$= \log \frac{\bar{q}_{X|U,Y}(x|u,y)}{\bar{q}_{X|Y}(x|y)} \tag{43}$$

$$= R_{\mathrm{WZ}}(D) - \frac{(x-\psi(u,y))^2}{2D} + \frac{(x-\rho y)^2}{2S}, \tag{44}$$

where (43) follows from Bayes' rule and $U - X - Y$, and (44) substitutes the Gaussian densities with conditional means $\psi(u, y)$ and ρy and variances D and S.

For block vectors, use the coordinatewise reconstruction

$$\psi(u^n, y^n) := (\psi(u_i, y_i))_{i\in[n]}. \tag{45}$$

For fixed (x^n, u^n) and $Z^n \sim \mathcal{N}(0_n, qI_n)$, define the conditional distortion mean $\bar{d}(x^n, u^n)$ by

$$\bar{d}(x^n,u^n) := \mathsf{E}_{Z^n}[\|x^n - \psi(u^n, x^n + Z^n)\|^2] \tag{46}$$

$$= \mathsf{E}_{Z^n}\left[\left\|\left(1-\frac{D}{q}\right)(x^n-u^n) - \frac{D}{q}Z^n\right\|^2\right] \tag{47}$$

$$= \left(1-\frac{D}{q}\right)^2 \|x^n-u^n\|^2 + \frac{nD^2}{q}, \tag{48}$$

where (47) substitutes (38), and (48) uses the zero mean and covariance qI_n of Z^n. Define the conditional information mean by

$$\bar{g}(x^n,u^n) := \mathsf{E}_{Z^n}\left[\sum_{i=1}^n g(x_i,u_i,x_i+Z_i)\right]. \tag{49}$$

Recalling the multiplier λ in (20) and the source-energy fluctuation A_n in (21), we obtain

$$\bar{g}(x^n,u^n) + \lambda\big(\bar{d}(x^n,u^n) - nD\big) = nR_{\mathrm{WZ}}(D) + \frac{\mathsf{E}_{Z^n}[\|(1-\rho)x^n - \rho Z^n\|^2]}{2S} - \frac{n}{2} \tag{50}$$

$$= nR_{\mathrm{WZ}}(D) + \frac{(1-\rho)^2\|x^n\|^2 + n\rho^2 q}{2S} - \frac{n}{2} \tag{51}$$

$$= nR_{\mathrm{WZ}}(D) + \frac{1-\rho}{2}\left(\frac{\|x^n\|^2}{p} - n\right) \tag{52}$$

$$= nR_{\mathrm{WZ}}(D) + \sqrt{n}A_n(x^n), \tag{53}$$

where (50) follows by summing (44) and cancelling the distortion terms, (51) averages the Gaussian noise, (52) uses the definitions of S and ρ in (13) and (9), and (53) recalls (21). Thus reducing the conditional distortion mean raises the conditional information mean by the corresponding amount scaled by λ.

B. Coding Scheme

We use the mean relation (53) to choose a conditional auxiliary distribution for each source sequence. Fix a source cutoff $b > 0$ and shell width $\delta > 0$. With L, δ, p, q, D fixed, $O_b(1)$ denotes a bounded quantity whose bound may depend on b, and $o_b(1)$ denotes a quantity tending to zero for each fixed b. Uniformity over the source or shell is stated at each use. Define the set of source sequences retained in the construction by

$$\mathcal{C}_{n,b} := \{x^n \in \mathbb{R}^n : |A_n(x^n)| \le b\}. \tag{54}$$

Recall the Borel maximizing selection $s^\star$ in Proposition 1. For $x^n \in \mathcal{C}_{n,b}$, choose the allocation

$$s(x^n) := s^\star(L - A_n(x^n)). \tag{55}$$

Define its squared shell radius by

$$r_n(x^n) := nT - \frac{\sqrt{n}\, s(x^n)}{\lambda(1-D/q)^2}. \tag{56}$$

The corresponding shell is

$$\mathcal{E}_n(x^n) := \{u^n \in \mathbb{R}^n : |\|x^n-u^n\|^2 - r_n(x^n)| \le \delta\}. \tag{57}$$

Boundedness of $s^\star$ on compact intervals ensures that $r_n(x^n) > 0$ uniformly over retained x^n for all sufficiently large n. Write $\bar{Q}_{U^n|X^n}$ for the product reference kernel $(\bar{Q}_{U|X})^{\otimes n}$, and define its shell probability

$$\alpha_n(x^n) := \bar{Q}_{U^n|X^n=x^n}(\mathcal{E}_n(x^n)). \tag{58}$$

For $x^n \in \mathcal{C}_{n,b}$, choose the source-dependent encoder kernel

$$R_{U^n|X^n=x^n}(du^n) := \frac{\mathbb{1}\{u^n \in \mathcal{E}_n(x^n)\}}{\alpha_n(x^n)} \bar{Q}_{U^n|X^n=x^n}(du^n). \tag{59}$$

Outside $\mathcal{C}_{n,b}$, use the unconditioned reference kernel. Choose the decoder reference

$$\bar{Q}_{U^n|Y^n=y^n} := (\bar{Q}_{U|Y})^{\otimes n}(\cdot|y^n). \tag{60}$$

The Borel selection $s^\star$ makes the shell and its probability measurable, so (59) defines a probability kernel. Both kernels have densities relative to $(\bar{Q}_U)^{\otimes n}$, and the Gaussian decoder reference has positive density everywhere. Thus the measurable-kernel and absolute-continuity hypotheses of the conditional Poisson matching lemma hold. Define the auxiliary joint distribution

$$\widehat{P} := P_{X^nY^n} R_{U^n|X^n} \tag{61}$$

and its information density relative to the decoder reference by

$$\imath_n(x^n, u^n, y^n) := \log \frac{dR_{U^n|X^n=x^n}}{d\bar{Q}_{U^n|Y^n=y^n}}(u^n). \tag{62}$$

The source and side-information marginal of $\widehat{P}$ is the original distribution $P_{X^nY^n}$. The conditional distribution of U^n given Y^n under $\widehat{P}$ need not equal the chosen decoder reference $\bar{Q}_{U^n|Y^n}$.

Applying the conditional Poisson matching lemma [9, Lemma 2] with encoder kernel $R_{U^n|X^n}$ and decoder reference $\bar{Q}_{U^n|Y^n}$, as in the proof of [9, Theorem 3], gives the following bound. For every positive integer M_n and every $\gamma > 0$, there exists a deterministic (n, M_n)-code satisfying

$$P_{\mathrm{e}}^{(n)}(D) \le 1 - \mathsf{E}_{\widehat{P}}\left[\frac{\mathbb{1}\{\|X^n - \psi(U^n, Y^n)\|^2 \le nD\}}{1 + \exp\{\imath_n(X^n, U^n, Y^n)\}/M_n}\right] \tag{63}$$

$$\le \widehat{P}\big\{\|X^n - \psi(U^n, Y^n)\|^2 > nD \text{ or } \imath_n(X^n, U^n, Y^n) > \log M_n - \gamma\big\} + e^{-\gamma}, \tag{64}$$

where the cited construction gives (63) with density ratio $\exp\{\imath_n(X^n, U^n, Y^n)\}/M_n$, and (64) restricts the expectation to the information and distortion constraints and uses $(1+t)^{-1} \ge 1 - t$ for $t \ge 0$. The cited construction transmits a label in $[M_n]$. Fixing a realization of the random code whose error does not exceed its average gives a deterministic encoder and decoder. For the finitely many initial blocklengths, use any fixed code.

C. Shell Bounds and Gaussian Approximation

The following lemma gives the shell probability, the squared-norm and inner-product limits, and the conditional means needed to evaluate (64).

Lemma 1. *Fix $b, \delta > 0$. There is a constant $c_b > 0$ such that, for all sufficiently large n,*

$$\inf_{x^n \in \mathcal{C}_{n,b}} \alpha_n(x^n) \ge c_b n^{-1/2}. \tag{65}$$

Under $\widehat{P}$, the normalized squared norms and inner product satisfy

$$\left(\frac{\|X^n\|^2}{n}, \frac{\|X^n - U^n\|^2}{n}, \frac{(X^n)^\top (X^n - U^n)}{n}\right) \longrightarrow (p, T, T) \tag{66}$$

in probability, uniformly conditionally on $X^n = x^n \in \mathcal{C}_{n,b}$. Explicitly, for every $\eta > 0$, the conditional probability that any component in (66) *differs from its stated limit by more than η tends to zero uniformly over retained x^n. The first two deviations converge to zero deterministically on the shell. Furthermore, uniformly over retained source-shell pairs,*

$$\bar{d}(x^n, u^n) = nD - \frac{\sqrt{n}}{\lambda} s(x^n) + O_b(1). \tag{67}$$

The corresponding information mean satisfies

$$\bar{g}(x^n, u^n) = nR_{\mathrm{WZ}}(D) + \sqrt{n}\big(A_n(x^n) + s(x^n)\big) + O_b(1). \tag{68}$$

The proof of Lemma 1 is provided in Appendix C. The lower bound on $\alpha_n(x^n)$ limits the information cost of conditioning to $\frac{1}{2}\log n - \log c_b$. This cost is negligible after division by $\sqrt{n}$, while the two conditional means give the allocation specified in (55).

We next evaluate the joint fluctuations in (64). The probability in that bound is under $\widehat{P}$, whose source and side-information marginal equals $P_{X^nY^n}$. The auxiliary reconstruction $\psi(U^n, Y^n)$ need not equal the reconstruction produced by the deterministic code.

Define the centered information score under $\widehat{P}$ by

$$J_n := \frac{\sum_{i=1}^n g(X_i, U_i, Y_i) - \bar{g}(X^n, U^n)}{\sqrt{n}}. \tag{69}$$

Define the scaled distortion score by

$$B_n := \frac{\lambda}{\sqrt{n}}\big(\|X^n - \psi(U^n, Y^n)\|^2 - \bar{d}(X^n, U^n)\big). \tag{70}$$

Conditional on (X^n, U^n), both scores fluctuate only through the observation noise $Z^n \sim \mathcal{N}(0_n, qI_n)$, independently of the source-description pair; no decoding event is imposed in this conditional distribution.

Lemma 2. *For the construction in Lemma 1,*

$$\sup_{x^n \in \mathcal{C}_{n,b}} \sup_{r,s \in \mathbb{R}} \big|\widehat{P}\{J_n \le r, B_n \le s \mid X^n = x^n\} - \Phi_\Sigma(r, s)\big| \longrightarrow 0, \tag{71}$$

where Σ is defined in (23)*. The Gaussian pair used in this limit can be taken independent of the source limit A.*

The proof of Lemma 2 is provided in Appendix D. We express J_n and B_n using two inner products with the Gaussian noise and its centered squared norm. Their joint limit, together with the finite-grid argument in [7, Appendix E-B, proof of Proposition 3], gives an approximation uniform over the source-dependent thresholds.

D. Final Steps

We now choose the message size and substitute Lemmas 1 and 2 into the error bound (64). Define

$$M_n := \left\lceil \exp\{nR_{\mathrm{WZ}}(D) + \sqrt{n}L + 3\log n\} \right\rceil. \tag{72}$$

Use the one-shot threshold

$$\gamma_n := \log n. \tag{73}$$

On the shell, the information density satisfies

$$\imath_n(x^n, u^n, y^n) = \log \frac{(\bar{q}_{U|X})^{\otimes n}(u^n|x^n)}{\alpha_n(x^n)(\bar{q}_{U|Y})^{\otimes n}(u^n|y^n)} \tag{74}$$

$$= \sum_{i=1}^n g(x_i, u_i, y_i) - \log \alpha_n(x^n), \tag{75}$$

where (74) follows from (62), (59), and (60), and (75) uses the product Gaussian densities and (42). The conditioning cost satisfies

$$-\log \alpha_n(x^n) \le -\log(c_b n^{-1/2}) \tag{76}$$

$$= \tfrac{1}{2}\log n - \log c_b, \tag{77}$$

where (76) follows from (65), and (77) expands the logarithm. For each fixed b, this cost is at most $2\log n$ for all sufficiently large n.

By the two mean expansions in Lemma 1, choose a constant $C_b > 0$ bounding both scaled mean remainders. For each $x^n \in \mathcal{C}_{n,b}$, define the event

$$\mathcal{A}_n(x^n) := \left\{ J_n \le L - A_n(x^n) - s(x^n) - \frac{C_b}{\sqrt{n}} \right\} \cap \left\{ B_n \le s(x^n) - \frac{C_b}{\sqrt{n}} \right\}. \tag{78}$$

Conditional on $X^n = x^n$, the event $\mathcal{A}_n(x^n)$ implies $\|X^n - \psi(U^n, Y^n)\|^2 \le nD$ by (67) and (70). The same event implies $\imath_n(X^n, U^n, Y^n) \le \log M_n - \gamma_n$ by (68), (69), (72), and (75)–(77). Its conditional probability satisfies

$$\widehat{P}\{\mathcal{A}_n(x^n) \mid X^n = x^n\} = \Phi_\Sigma\left(L - A_n(x^n) - s(x^n) - \frac{C_b}{\sqrt{n}}, s(x^n) - \frac{C_b}{\sqrt{n}}\right) + o_b(1) \tag{79}$$

$$= \Phi_\Sigma\big(L - A_n(x^n) - s(x^n), s(x^n)\big) + o_b(1) \tag{80}$$

$$= H^\star(L - A_n(x^n)) + o_b(1), \tag{81}$$

where (79) follows from Lemma 2, (80) uses the uniform continuity supplied by the continuous Gaussian marginals, and (81) follows from (55). Every remainder is uniform over $x^n \in \mathcal{C}_{n,b}$.

Combining the one-shot bound (64) with the inclusion for $\mathcal{A}_n(x^n)$ and the estimate (81) gives

$$P_{\mathrm{e}}^{(n)}(D) \le 1 - \widehat{P}\{\|X^n - \psi(U^n, Y^n)\|^2 \le nD, \imath_n(X^n, U^n, Y^n) \le \log M_n - \gamma_n\} + n^{-1} \tag{82}$$

$$\le 1 - \mathsf{E}_P\big[\mathbb{1}\{X^n \in \mathcal{C}_{n,b}\}\widehat{P}\{\mathcal{A}_n(X^n) \mid X^n\}\big] + n^{-1} \tag{83}$$

$$\le 1 - \mathsf{E}_P\big[\mathbb{1}\{|A_n(X^n)| \le b\}H^\star(L - A_n(X^n))\big] + o_b(1), \tag{84}$$

where (82) follows from (64) with (73), (83) retains the source cutoff and uses the favorable-rectangle inclusion, and (84) follows from (81). Passing to the limit gives

$$\limsup_{n\to\infty} P_{\mathrm{e}}^{(n)}(D) \le 1 - \mathsf{E}[\mathbb{1}\{|A| \le b\}H^\star(L - A)], \tag{85}$$

where (85) uses the source central limit theorem (22), boundedness and continuity of $H^\star$, and the zero probability of the Gaussian cutoff endpoints. The allocation is selected before the noise is realized, so the optimization defining $H^\star$ occurs inside the source expectation. The source-dependent kernel (59) is part of the fixed encoder, and only its selected index M is transmitted.

Finally, we let the source cutoff increase with the blocklength. For each integer $j \ge 1$, use the construction with $b = j$ and fixed δ. Choose increasing blocklength thresholds n_j such that, for every $n \ge n_j$, the error of the jth construction is at most the right-hand side of (85) with $b = j$, plus j^{-1}. Also choose n_j large enough that the conditioning cost (77) fits within the overhead in (72). Use the jth construction for $n_j \le n < n_{j+1}$. It follows that

$$\limsup_{n\to\infty} P_{\mathrm{e}}^{(n)}(D) \le 1 - \lim_{j\to\infty} \mathsf{E}[\mathbb{1}\{|A| \le j\}H^\star(L - A)] \tag{86}$$

$$= 1 - \mathsf{E}[H^\star(L - A)] \tag{87}$$

$$= 1 - F_G(L), \tag{88}$$

where (86) follows from the choice of n_j, (87) uses monotone convergence, and (88) recalls (26). The message size remains (72), and its integer rounding costs $O(1)$ because $R_{\mathrm{WZ}}(D) > 0$. This proves (35) and (36).

Set $L = L_G(D, \varepsilon)$. By Proposition 1, $F_G(L_G(D, \varepsilon)) = 1 - \varepsilon$. Hence

$$L^\star_{\mathrm{WZ}}(D, \varepsilon) \le L_G(D, \varepsilon), \tag{89}$$

where (89) follows from Definition 2, (35), and (36), since $\log n/\sqrt{n} \to 0$. This completes the achievability proof of Theorem 1.

V. Converse Proof

Fix $L \in \mathbb{R}$ and any sequence of (n, M_n)-codes satisfying

$$\log M_n \le nR_{\mathrm{WZ}}(D) + \sqrt{n}L + o(\sqrt{n}). \tag{90}$$

Define the non-excess-distortion event by

$$\mathcal{S}_n := \{\|X^n - \hat{X}^n\|^2 \le nD\}. \tag{91}$$

We prove the converse part of Theorem 1 by showing that

$$\limsup_{n\to\infty} P(\mathcal{S}_n) \le F_G(L). \tag{92}$$

We first select source sequences on which the conditional success probability is not too small, and obtain bounds on the posterior mean and covariance after adding a Gaussian observation. We next compare two expressions for the message likelihood and identify an allocation determined by the source and an independent permutation. Finally, a conditional Gaussian approximation and averaging under the original source distribution give (92). The proof follows the source-dependent reduction in [7, Secs. IV-B–IV-D], with the reconstruction, posterior bounds, and likelihood comparisons proved below for the Gaussian model.

Throughout this proof, P denotes the joint distribution induced by the Gaussian source and channel in (1), the code $M = f(X^n)$ and $\hat{X}^n = \phi(M, Y^n)$, and the independent auxiliary randomness introduced below. It suffices to consider a subsequence on which $P(\mathcal{S}_n) \ge c_0$ for some constant $c_0 > 0$. All asymptotic estimates below concern this fixed code sequence and subsequence.

A. Source Selection and Posterior Bounds

Write $\|x^n\|_\infty := \max_{i\in[n]} |x_i|$. For each $x^n \in \mathbb{R}^n$, define its conditional success probability $s_n(x^n) := P(\mathcal{S}_n \mid X^n = x^n)$ and the indicator for retaining that source sequence:

$$F_n(x^n) := \mathbb{1}\{s_n(x^n) \geq n^{-1},\ \|x^n\|^2 \leq 2pn,\ \|x^n\|_\infty \leq \sqrt{8p\log n}\}. \tag{93}$$

Write $F_n := F_n(X^n)$ and $\beta_n := P(F_n = 1)$. Define the distribution conditioned on the retained source set by

$$\tilde{P}(\mathcal{E}) := \frac{P(\mathcal{E} \cap \{F_n = 1\})}{\beta_n} \tag{94}$$

for every measurable event $\mathcal{E}$. For some constant $c_{\mathrm{t}} > 0$, we obtain

$$P(\mathcal{S}_n, F_n = 0) \leq \mathsf{E}_P[s_n(X^n)\mathbb{1}\{s_n(X^n) < n^{-1}\}] + P(\|X^n\|^2 > 2pn) + P(\|X^n\|_\infty > \sqrt{8p\log n}) \tag{95}$$

$$\leq n^{-1} + e^{-c_{\mathrm{t}} n} + 2n^{-3} \tag{96}$$

$$= o(1), \tag{97}$$

where (95) follows from conditioning on X^n and the union bound, (96) uses the flag threshold, a chi-square upper-tail bound, and the coordinatewise Gaussian tail bound, and (97) follows by taking $n \to \infty$. Thus β_n stays bounded away from zero on the selected subsequence. For every event $\mathcal{E}$,

$$P(\mathcal{E}, F_n = 1) = \beta_n \tilde{P}(\mathcal{E}), \tag{98}$$

where (98) uses the definition of $\tilde{P}$ in (94). Moreover, for every retained source vector x^n,

$$\tilde{P}_{Y^n|X^n=x^n} = \mathcal{N}(x^n, qI_n), \tag{99}$$

where (99) follows because F_n is determined by X^n. Thus, conditioning on $F_n = 1$ preserves the observation channel.

Let Π be an independent uniform permutation of $[n]$. Set $k := \lceil n^{3/4} \rceil$, $\Delta_n := n^{-2}$, and $h_n := \sqrt{8p\log n}$. For $x \in \mathbb{R}$, define the finite quantizer

$$\chi_n(x) := \begin{cases} -1, & \text{if } x < -2h_n, \\ \lfloor (x + 2h_n)/\Delta_n \rfloor, & \text{if } -2h_n \leq x < 2h_n, \\ \lceil 4h_n/\Delta_n \rceil, & \text{if } x \geq 2h_n, \end{cases} \tag{100}$$

where the first and last values label the two overflow intervals. Define the sampled description $V^{\mathrm{pre}} := (\chi_n(X_{\Pi(i)}))_{i\in[k]}$, the augmented message $M_0 := (M, F_n)$, and $W := (M_0, V^{\mathrm{pre}})$. For each permutation π, let $\mathcal{W}_\pi$ be the range of W, and define its source cells by $\mathcal{C}_{w,\pi} := \{x^n : W(x^n, \pi) = w\}$ for $w \in \mathcal{W}_\pi$. Then

$$\log |\mathcal{W}_\pi| \leq \log(2M_n) + O(k \log n), \tag{101}$$

where (101) uses the $O(n^2\sqrt{\log n})$ possible values of the quantizer in (100). The sampled description V^{pre} permits selection of a reconstruction with small average distortion. Its description cost cannot be discarded on the $\sqrt{n}$ scale and will be retained in both likelihood comparisons.

For $t \geq t_0 := q^{-1}$, define $O_t^n := \sqrt{t}X^n + G^n$, where $G^n \sim \mathcal{N}(0_n, I_n)$ is independent of (X^n, W, Π), and let $D_t := (D^{-1} + t - t_0)^{-1}$. The conditional distribution $\tilde{P}_{X^n|W,O_t^n,\Pi}$ is obtained by restricting the source distribution in (94) to $\mathcal{C}_{w,\pi}$ and applying the Gaussian likelihood $O_t^n \mid X^n = x^n \sim \mathcal{N}(\sqrt{t}x^n, I_n)$. No conditioning on $\mathcal{S}_n$ is imposed. Define its mean $m_t^n := \mathsf{E}_{\tilde{P}}[X^n \mid W, O_t^n, \Pi]$ and covariance $V_t := \mathrm{Cov}_{\tilde{P}}(X^n \mid W, O_t^n, \Pi)$. To compare the posterior with a Gaussian distribution, define

$$K(t) := \mathsf{E}_{\tilde{P}}\big[\,\mathrm{KL}(\tilde{P}_{X^n|W,O_t^n,\Pi} \| \mathcal{N}(m_t^n, D_t I_n))\big]. \tag{102}$$

Recall that $\|C\|_{\mathrm{F}}^2 = \mathrm{tr}(C^\top C)$ for a matrix C.

Lemma 3. *There is a deterministic $\nu_n \in (0, n^{-1/32}]$ such that*

$$K(t_0 + \nu_n) = o(n) \tag{103}$$

and

$$\mathsf{E}_{\tilde{P}}[\|V_{t_0+\nu_n} - D_{t_0+\nu_n} I_n\|_{\mathrm{F}}^2] = o(n). \tag{104}$$

The proof of Lemma 3 is provided in Appendix E. The relative entropy bound will determine the squared estimation error, while the covariance bound will control the likelihood fluctuation. To obtain these bounds, we use V^{pre} to select a reconstruction with average squared error $nD + O(n^{15/16})$, which gives $K(t_0) = O(n^{15/16})$. The conditional entropy derivative [11, Theorem 2] and the derivative of the mean squared estimation error [12, Corollary 1, Eq. (45)] then give

$$K'(t) = -\frac{1}{2D_t}\mathsf{E}_{\tilde{P}}[\|V_t - D_t I_n\|_{\mathrm{F}}^2], \tag{105}$$

where (105) follows by differentiating (102) using the two cited identities; the full calculation is given in (329)–(332). Integrating over $[t_0, t_0 + n^{-1/32}]$ gives the required deterministic ν_n. The value of ν_n may depend on the fixed code sequence and $\tilde{P}$, but not on the realized source, permutation, or noise.

Set $\nu := \nu_n$ and introduce $A^{n,\mathrm{ext}} := \sqrt{\nu} X^n + G^{n,\mathrm{ext}}$, where $G^{n,\mathrm{ext}} \sim \mathcal{N}(0_n, I_n)$ is independent of all preceding variables. Define the enhanced noise variance by

$$\tilde{q} := (q^{-1} + \nu)^{-1} \tag{106}$$

and the enhanced noise vector by

$$\tilde{Z}^n := \tilde{q}(q^{-1} Z^n + \sqrt{\nu} G^{n,\mathrm{ext}}). \tag{107}$$

The enhanced observation satisfies

$$\begin{aligned} \tilde{Y}^n &:= \tilde{q}\big(q^{-1} Y^n + \sqrt{\nu} A^{n,\mathrm{ext}}\big) && (108)\\ &= \tilde{q}\big((q^{-1} + \nu) X^n + q^{-1} Z^n + \sqrt{\nu} G^{n,\mathrm{ext}}\big) && (109)\\ &= X^n + \tilde{Z}^n, && (110) \end{aligned}$$

where (108) defines $\tilde{Y}^n$, (109) uses $Y^n = X^n + Z^n$ in (1) and the definition of $A^{n,\mathrm{ext}}$, and (110) uses (106) and (107). Thus $\tilde{Z}^n \sim \mathcal{N}(0_n, \tilde{q} I_n)$ and is independent of (X^n, Π) under both P and $\tilde{P}$.

Let $\tilde{S} := S/(1 + \nu S)$, $\tilde{D} := D/(1 + \nu D)$, and $\tilde{\rho} := p/(p + \tilde{q})$. The corresponding first-order rate is

$$\begin{aligned} \tilde{R} &:= \frac{1}{2} \log \frac{\tilde{S}}{\tilde{D}} && (111)\\ &= \frac{1}{2} \log \frac{S}{D} - \frac{1}{2} \log \frac{1 + \nu S}{1 + \nu D} && (112)\\ &= R_{\mathrm{WZ}}(D) - \frac{1}{2} \log \frac{1 + \nu S}{1 + \nu D}, && (113) \end{aligned}$$

where (111) defines $\tilde{R}$, (112) substitutes $\tilde{S}$ and $\tilde{D}$, and (113) uses (15). Since $\tilde{Y}^n/\sqrt{\tilde{q}}$ has the same conditional distribution as $O^n_{t_0+\nu}$ and $\tilde{D} = D_{t_0+\nu}$, Lemma 3 applies to the posterior given $(W, \tilde{Y}^n, \Pi)$. Its relative entropy from the Gaussian reference with covariance $\tilde{D} I_n$ and the posterior mean is $o(n)$ on average, and the expected squared Frobenius norm of the covariance difference is $o(n)$. The original success probability satisfies

$$\begin{aligned} P(\mathcal{S}_n) &= P(\mathcal{S}_n, F_n = 1) + o(1) && (114)\\ &= \beta_n \tilde{P}(\mathcal{S}_n) + o(1), && (115) \end{aligned}$$

where (114) uses (97), and (115) uses (98). We next use the posterior bounds to control $\beta_n \tilde{P}(\mathcal{S}_n)$.

B. Likelihood Comparisons and Source-Dependent Allocation

For $v > 0$ and a positive integer d, let φ_{vI_d} denote the density of $\mathcal{N}(0_d, vI_d)$. Under the original distribution P, $X^n \mid \tilde{Y}^n = y^n \sim \mathcal{N}(\tilde{\rho} y^n, \tilde{S} I_n)$. Therefore the conditional mass function of W is

$$P_{W|\tilde{Y}^n,\Pi}(w \mid y^n, \pi) = \int_{\mathcal{C}_{w,\pi}} \varphi_{\tilde{S} I_n}(x^n - \tilde{\rho} y^n)\, dx^n, \tag{116}$$

where (116) integrates the conditional Gaussian density over $\mathcal{C}_{w,\pi}$, defined above (101). Define the negative logarithm of the conditional message probability by

$$\tilde{H} := -\log P_{W|\tilde{Y}^n,\Pi}(W \mid \tilde{Y}^n, \Pi). \tag{117}$$

Since F_n is a coordinate of W, conditioning on a cell with $F_n = 1$ gives the same source distribution under P and $\tilde{P}$. However, the message probability in (116), and hence $\tilde{H}$, is always normalized under P. Define the deterministic cost associated with the sampled description by

$$b_n^{\mathrm{pre}} := k \left(\log \frac{1}{\Delta_n} + \frac{1}{2} \log(2\pi e \tilde{D}) \right). \tag{118}$$

Lemma 4. *For every $\eta > 0$,*

$$P\{\mathcal{S}_n, F_n = 1,\ \tilde{H} > n\tilde{R} + b_n^{\mathrm{pre}} + \sqrt{n}(L + \eta)\} \longrightarrow 0 \tag{119}$$

and

$$P\left\{\mathcal{S}_n, F_n = 1,\ \tilde{H} < n\tilde{R} + b_n^{\mathrm{pre}} + \frac{\|X^n - \tilde{\rho}\tilde{Y}^n\|^2 - n\tilde{S}}{2\tilde{S}} - \eta\sqrt{n}\right\} \longrightarrow 0. \tag{120}$$

The proof of Lemma 4 is provided in Appendix F. The upper bound includes the exact rate change caused by the added observation in (113). The lower bound uses a Gaussian reference centered at a reconstruction formed from the original decoder

and the added observation. Both bounds contain the same cost b_n^{pre}. We next separate the conditional mean of $\tilde{H}$ given the source and permutation from its fluctuation due to observation noise.

Recall $\tilde{P}$ in (94). Define the posterior mean of the source by $m^n := \mathsf{E}_{\tilde{P}}[X^n \mid W, \tilde{Y}^n, \Pi]$ and the vector

$$b^n := X^n - \mathsf{E}_{\tilde{P}}[m^n \mid X^n, \Pi]. \tag{121}$$

Define a second vector by

$$\ell^n := -\frac{X^n}{p+\tilde{q}} + \frac{b^n}{\tilde{q}}. \tag{122}$$

Recall the source-energy fluctuation A_n in (21). Define its enhanced counterpart by

$$\tilde{A}_n := \frac{1-\tilde{\rho}}{2\sqrt{n}} \left(\frac{\|X^n\|^2}{p} - n \right). \tag{123}$$

The vectors b^n and ℓ^n are determined by (X^n, Π). Define the two fluctuations due to $\tilde{Z}^n$ by

$$\tilde{J}_n := \frac{(\ell^n)^\top \tilde{Z}^n}{\sqrt{n}} + \frac{\tilde{\rho} - \tilde{D}/\tilde{q}}{2\tilde{q}\sqrt{n}} (\|\tilde{Z}^n\|^2 - n\tilde{q}) \tag{124}$$

and

$$\tilde{B}_n := -\frac{(b^n)^\top \tilde{Z}^n}{\tilde{q}\sqrt{n}} + \frac{\tilde{D}}{2\tilde{q}^2\sqrt{n}} (\|\tilde{Z}^n\|^2 - n\tilde{q}). \tag{125}$$

Lemma 5. *For every $\eta > 0$, the centered likelihood satisfies*

$$\tilde{P}\left\{ \left| \tilde{H} - \mathsf{E}_{\tilde{P}}[\tilde{H} \mid X^n, \Pi] - \sqrt{n}\tilde{J}_n \right| > \eta\sqrt{n} \right\} \longrightarrow 0. \tag{126}$$

Moreover, in $\tilde{P}$-probability,

$$\frac{\|X^n\|^2}{n} \longrightarrow p, \tag{127}$$

$$\frac{(X^n)^\top b^n}{n} - \tilde{D} \longrightarrow 0, \tag{128}$$

and

$$\frac{\|b^n\|^2}{n} - \left(\tilde{D} - \frac{\tilde{D}^2}{\tilde{q}} \right) \longrightarrow 0. \tag{129}$$

The proof of Lemma 5 is provided in Appendix G. We apply the Gaussian Poincaré inequality twice to bound the error in the likelihood approximation by the posterior covariance difference in (104). We then use the posterior relative entropy and Gaussian tail bounds to prove the squared-norm and inner-product limits in (127)–(129).

Define the allocation of the arbitrary code by

$$T_n := \frac{\mathsf{E}_{\tilde{P}}[\tilde{H} \mid X^n, \Pi] - n\tilde{R} - b_n^{\mathrm{pre}}}{\sqrt{n}} - \tilde{A}_n. \tag{130}$$

The allocation T_n is determined by (X^n, Π) and does not depend on the realized observation noise. For the normalized squared norm of $X^n - \tilde{\rho}\tilde{Y}^n$, we obtain

$$\frac{\|X^n - \tilde{\rho}\tilde{Y}^n\|^2 - n\tilde{S}}{2\tilde{S}\sqrt{n}} = \frac{\|(1-\tilde{\rho})X^n - \tilde{\rho}\tilde{Z}^n\|^2 - n\tilde{S}}{2\tilde{S}\sqrt{n}} \tag{131}$$

$$= \tilde{A}_n - \frac{(X^n)^\top \tilde{Z}^n}{(p+\tilde{q})\sqrt{n}} + \frac{\tilde{\rho}}{2\tilde{q}\sqrt{n}} (\|\tilde{Z}^n\|^2 - n\tilde{q}) \tag{132}$$

$$= \tilde{A}_n + \tilde{J}_n + \tilde{B}_n, \tag{133}$$

where (131) uses (110), (132) expands the square and uses $\tilde{S} = p(1-\tilde{\rho}) = \tilde{q}\tilde{\rho}$, and (133) follows from (122)–(125). For any fixed $\eta > 0$, substituting the likelihood decomposition in (126) and the allocation in (130) into (119) gives

$$P\{\mathcal{S}_n, F_n = 1,\ \tilde{J}_n > L - \tilde{A}_n - T_n + \eta\} \longrightarrow 0. \tag{134}$$

Similarly, (120) and (133) give

$$P\{\mathcal{S}_n, F_n = 1,\ \tilde{B}_n > T_n + \eta\} \longrightarrow 0. \tag{135}$$

In both substitutions, (98) transfers the vanishing probability in (126) to the original distribution, and the terms $n\tilde{R}$ and b_n^{pre} cancel. Consequently,

$$P(\mathcal{S}_n) \le P\{F_n = 1,\ \tilde{J}_n \le L - \tilde{A}_n - T_n + \eta,\ \tilde{B}_n \le T_n + \eta\} + o(1), \tag{136}$$

where (136) follows from (97), (134), and (135) by the union bound. We next bound the joint probability of the two constraints on $(\tilde{J}_n, \tilde{B}_n)$ in (136), conditional on (X^n, Π).

C. Gaussian Bound on the Non-Excess-Distortion Probability

Conditioned on (X^n, Π), the vectors b^n and ℓ^n are fixed and $\tilde{Z}^n$ is Gaussian. Consequently,

$$\mathrm{Var}_{\tilde{P}}(\tilde{J}_n \mid X^n, \Pi) = \frac{\tilde{q}\|\ell^n\|^2}{n} + \frac{1}{2}(\tilde{\rho} - \tilde{D}/\tilde{q})^2, \tag{137}$$

$$\mathrm{Var}_{\tilde{P}}(\tilde{B}_n \mid X^n, \Pi) = \frac{\|b^n\|^2}{\tilde{q}n} + \frac{1}{2}(\tilde{D}/\tilde{q})^2, \tag{138}$$

and

$$\mathrm{Cov}_{\tilde{P}}(\tilde{J}_n, \tilde{B}_n \mid X^n, \Pi) = -\frac{(\ell^n)^\top b^n}{n} + \frac{1}{2}(\tilde{\rho} - \tilde{D}/\tilde{q})\frac{\tilde{D}}{\tilde{q}}, \tag{139}$$

where (137)–(139) use the vanishing odd Gaussian moments and $\mathrm{Var}(\|\tilde{Z}^n\|^2) = 2n\tilde{q}^2$. Substituting (122) and (127)–(129), and using $\nu_n \to 0$, shows that the conditional covariance of $(\tilde{J}_n, \tilde{B}_n)$ converges in $\tilde{P}$-probability to Σ in (23).

The Gaussian limit follows directly from the exact conditional characteristic function. Fix $(v_1, v_2) \in \mathbb{R}^2$, write $\mathrm{i} := \sqrt{-1}$, and define $c := v_1(\tilde{\rho} - \tilde{D}/\tilde{q}) + v_2\tilde{D}/\tilde{q}$ and the vector $d_0^n := v_1\ell^n - v_2 b^n/\tilde{q}$. Then

$$\mathsf{E}_{\tilde{P}}[e^{\mathrm{i}(v_1\tilde{J}_n + v_2\tilde{B}_n)} \mid X^n, \Pi] = \mathsf{E}_{\tilde{P}}\left[\exp\left\{\frac{\mathrm{i}(d_0^n)^\top \tilde{Z}^n}{\sqrt{n}} + \frac{\mathrm{i}c(\|\tilde{Z}^n\|^2 - n\tilde{q})}{2\tilde{q}\sqrt{n}}\right\} \middle| X^n, \Pi\right] \tag{140}$$

$$= e^{-\mathrm{i}c\sqrt{n}/2}(1 - \mathrm{i}c/\sqrt{n})^{-n/2} \exp\left\{-\frac{\tilde{q}\|d_0^n\|^2}{2n(1 - \mathrm{i}c/\sqrt{n})}\right\} \tag{141}$$

$$= \exp\left\{-\frac{c^2}{4} - \frac{\tilde{q}\|d_0^n\|^2}{2n} + O\left(\frac{1 + \tilde{q}\|d_0^n\|^2/n}{\sqrt{n}}\right)\right\}, \tag{142}$$

where (140) uses the definitions of $\tilde{J}_n$ and $\tilde{B}_n$ in (124) and (125), together with the preceding definitions of c and d_0^n, (141) evaluates the Gaussian integral by completing the square, and (142) expands $\log(1 - \mathrm{i}c/\sqrt{n})$ and its reciprocal. The remainder is uniform when the normalized squared norms and inner product in (127)–(129) remain bounded.

Choose deterministic tolerances tending to zero sufficiently slowly. Let $\mathcal{G}_n$ consist of the pairs (X^n, Π) for which $F_n = 1$ and the three errors in (127)–(129) are within those tolerances. Lemma 5 permits this choice with $\tilde{P}(\mathcal{G}_n) \to 1$. Since $\mathcal{G}_n$ depends only on (X^n, Π), the conditional distribution of $\tilde{Z}^n$ remains Gaussian. Equations (137)–(142) give bounded conditional second moments and Gaussian convergence along every deterministic sequence in $\mathcal{G}_n$. Therefore,

$$\sup_{(x^n,\pi)\in\mathcal{G}_n} \sup_{r,s\in\mathbb{R}} \left|P(\tilde{J}_n \le r, \tilde{B}_n \le s \mid X^n = x^n, \Pi = \pi) - \Phi_\Sigma(r, s)\right| \longrightarrow 0, \tag{143}$$

where (143) follows by the finite-grid argument in [7, Appendix E-B, proof of Proposition 3]: the Gaussian marginals are continuous, convergence holds along every sequence in $\mathcal{G}_n$, and the bounded second moments control the tails. The conditional noise distribution is unchanged by conditioning on $F_n = 1$, as in (99). Since (143) is uniform over all thresholds (r, s), no boundedness assumption on T_n is required.

D. Final Steps

Fix $\eta > 0$. We combine (136) with the uniform Gaussian approximation (143) to obtain

$$1 - P_{\mathrm{e}}^{(n)}(D) = P(\mathcal{S}_n) \tag{144}$$

$$\le P(F_n = 1, \mathcal{G}_n, \tilde{J}_n \le L - \tilde{A}_n - T_n + \eta, \tilde{B}_n \le T_n + \eta) + o(1) \tag{145}$$

$$\le \mathsf{E}_P\big[\mathbb{1}\{F_n = 1, \mathcal{G}_n\}\Phi_\Sigma(L - \tilde{A}_n - T_n + \eta, T_n + \eta)\big] + o(1) \tag{146}$$

$$\le \mathsf{E}_P[H^\star(L - \tilde{A}_n + 2\eta)] + o(1), \tag{147}$$

where (144) uses (6) and (91), (145) uses (136), $\tilde{P}(\mathcal{G}_n^c) = o(1)$, and (98), (146) conditions on (X^n, Π) and applies (143), and (147) takes the supremum over the allocation and removes the nonnegative indicator. The conditional Gaussian approximation is applied without conditioning on $\mathcal{S}_n$.

Recall the source central limit theorem (22), which holds under the original distribution P. Since $\tilde{\rho} \to \rho$, the enhanced score $\tilde{A}_n$ in (123) has the same limit A. Hence

$$\limsup_{n\to\infty} P(\mathcal{S}_n) \le \mathsf{E}[H^\star(L - A + 2\eta)], \tag{148}$$

where (148) follows from (147) and the bounded continuity of $H^\star$ proved in Appendix A. Furthermore,

$$\lim_{\eta\downarrow 0} \mathsf{E}[H^\star(L - A + 2\eta)] = \mathsf{E}[H^\star(L - A)] \tag{149}$$

$$= F_G(L), \tag{150}$$

where (149) follows from bounded convergence and (150) is (26). Letting $\eta \downarrow 0$ in (148) therefore gives

$$\limsup_{n\to\infty} P(\mathcal{S}_n) \le F_G(L). \tag{151}$$

This proves (92); subsequences with vanishing success probability satisfy it directly. By Proposition 1, $L < L_G(D,\varepsilon)$ implies $F_G(L) < 1-\varepsilon$, so such a rate is not achievable under Definition 2. Combining this converse with (89) completes the proof of Theorem 1.

VI. Conclusion

In this paper, we derived the exact second-order coding rate for the Gaussian Wyner–Ziv problem under quadratic distortion and an excess-distortion probability criterion, for fixed positive variances and distortion strictly between zero and the conditional source variance. A source-dependent Gaussian shell construction and a converse for arbitrary measurable codes yield the same optimized joint information-distortion probability. The characterization shows a strict second-order rate loss relative to both-terminal side information, despite equality of the first-order rates.

We next discuss future directions. Firstly, the present result assumes a fixed interior distortion. Characterizing the behavior when the distortion approaches zero or the conditional source variance requires control of the estimates near those endpoints. Secondly, extending the analysis to vector Gaussian sources would permit quadratic distortion constraints on correlated components. Finally, quantitative bounds on the posterior approximation and conditional Gaussian convergence would be needed to investigate third-order asymptotics.

Appendix A
Proof of Proposition 1

Recall Σ in (23), $H^\star$ in (25), and F_G in (26). We first show that the maximizing allocation exists and can be chosen measurably. We then prove continuity, strict increase, and the endpoint limits of F_G. Write Σ_{ij} for the (i,j) entry of Σ. Since $0 < D/q < \rho < 1$, the determinant satisfies

$$\begin{aligned}
\det\Sigma &= \Sigma_{11}\Sigma_{22} - \Sigma_{12}^2 && (152)\\
&= \frac{D}{q}\left(\rho - \frac{D}{q}\right)\left[\left(1 - \frac{\rho - D/q}{2}\right)\left(1 - \frac{D}{2q}\right) - \frac{D}{4q}\left(\rho - \frac{D}{q}\right)\right] && (153)\\
&= \frac{D}{q}\left(\rho - \frac{D}{q}\right)\left(1 - \frac{\rho}{2}\right) && (154)\\
&> 0, && (155)
\end{aligned}$$

where (152) is the determinant formula for a symmetric 2×2 matrix, (153) substitutes (23), (154) cancels the two terms containing $(D/q)(\rho - D/q)$, and (155) uses $0 < D/q < \rho < 1$. The same parameter bounds make both diagonal entries positive. Thus Σ is positive definite.

For $(z,s) \in \mathbb{R}^2$, define

$$\Psi(z,s) := \Phi_\Sigma(z-s, s). \tag{156}$$

Fix $b > 0$ and restrict z to $[-b,b]$. Continuity and positivity of the Gaussian CDF with positive-definite covariance imply

$$\begin{aligned}
\eta_b &:= \min_{|z|\le b} \Psi(z,0) && (157)\\
&> 0, && (158)
\end{aligned}$$

where (158) follows because a positive continuous function has a positive minimum on a compact interval. For $\tau > b$ and $|z| \le b$, if $s \le -\tau$, then $\Psi(z,s) \le \mathbb{P}\{B \le -\tau\}$; if $s \ge \tau$, then $\Psi(z,s) \le \mathbb{P}\{J \le b - \tau\}$. Consequently,

$$\begin{aligned}
\limsup_{\tau\to\infty} \sup_{\substack{|z|\le b\\ |s|\ge\tau}} \Psi(z,s) &\le \lim_{\tau\to\infty} \big(\mathbb{P}\{B \le -\tau\} + \mathbb{P}\{J \le b - \tau\}\big) && (159)\\
&= 0, && (160)
\end{aligned}$$

where (159) uses the two marginal bounds, and (160) follows from Gaussian tail convergence. Choose $\tau_b > b$ so that the supremum outside $[-\tau_b, \tau_b]$ is smaller than η_b. Since $s = 0$ achieves at least η_b, every maximizer for $|z| \le b$ belongs to $[-\tau_b, \tau_b]$. The maximum is attained there by continuity on a compact interval.

We next prove continuity of $H^\star$. Fix any sequence $z_m \to z$, and choose a maximizer s_0 at z. Then

$$\begin{aligned}
\liminf_{m\to\infty} H^\star(z_m) &\ge \lim_{m\to\infty} \Psi(z_m, s_0) && (161)\\
&= \Psi(z, s_0) && (162)\\
&= H^\star(z), && (163)
\end{aligned}$$

where (161) evaluates the supremum at s_0, (162) uses continuity of Ψ, and (163) uses the maximizing choice of s_0. To bound the upper limit, choose a subsequence z_{m_j} along which $H^\star(z_{m_j})$ converges to that upper limit, and choose maximizers s_j there. The compact bound just proved permits a further subsequence with $s_j \to s_\infty$. Along this subsequence,

$$\limsup_{m\to\infty} H^\star(z_m) = \lim_{j\to\infty} \Psi(z_{m_j}, s_j) \tag{164}$$
$$= \Psi(z, s_\infty) \tag{165}$$
$$\le H^\star(z), \tag{166}$$

where (164) follows from the choices of the subsequence and its maximizers, (165) uses joint continuity, and (166) is the supremum property. Combining (163) and (166) proves continuity of $H^\star$.

Define $s^\star(z)$ to be the smallest maximizer at z, which exists because the set of maximizers is nonempty and compact. For any sequence $z_m \to z$, choose a subsequence on which $s^\star(z_m)$ converges to its lower limit, denoted by s_∞. The compact bound on the maximizers ensures that s_∞ is finite. We obtain

$$\Psi(z, s_\infty) = \lim_{j\to\infty} \Psi(z_{m_j}, s^\star(z_{m_j})) \tag{167}$$
$$= \lim_{j\to\infty} H^\star(z_{m_j}) \tag{168}$$
$$= H^\star(z), \tag{169}$$

where (167) follows from joint continuity, (168) uses the maximizing selections, and (169) uses continuity of $H^\star$. Thus s_∞ is a maximizer at z, and its value is at least the smallest such maximizer. Hence

$$\liminf_{m\to\infty} s^\star(z_m) \ge s^\star(z), \tag{170}$$

where (170) uses (169) and the definition of $s^\star(z)$ as the smallest maximizer. This proves lower semicontinuity, so $s^\star$ is Borel measurable and bounded on compact intervals.

To prove that $H^\star$ is strictly increasing, fix $z_2 > z_1$ and let $s_1 := s^\star(z_1)$. Then

$$H^\star(z_2) \ge \Psi(z_2, s_1) \tag{171}$$
$$= \Psi(z_1, s_1) + \mathbb{P}\{z_1 - s_1 < J \le z_2 - s_1,\ B \le s_1\} \tag{172}$$
$$> \Psi(z_1, s_1) \tag{173}$$
$$= H^\star(z_1), \tag{174}$$

where (171) evaluates the supremum at s_1, (172) decomposes the enlarged rectangle, (173) uses the strictly positive Gaussian density on the added strip, and (174) uses the maximizing choice at z_1.

For every $s \in \mathbb{R}$, the event $\{J \le z - s, B \le s\}$ is contained in $\{J + B \le z\}$. Conversely, choosing $s = z/2$ gives a lower bound on $H^\star(z)$. Hence,

$$\lim_{z\to-\infty} H^\star(z) \le \lim_{z\to-\infty} \mathbb{P}\{J + B \le z\} \tag{175}$$
$$= 0, \tag{176}$$

where (175) uses the half-space inclusion and (176) uses the Gaussian lower tail. Similarly,

$$\lim_{z\to\infty} H^\star(z) \ge \lim_{z\to\infty} \Phi_\Sigma(z/2, z/2) \tag{177}$$
$$= 1, \tag{178}$$

where (177) evaluates the feasible allocation $z/2$, and (178) follows because both Gaussian upper tails vanish. Since $H^\star$ takes values in $[0, 1]$, these bounds identify its endpoint limits.

Finally, fix any sequence $L_m \to L$. For the Gaussian source fluctuation A defined in Section III,

$$\lim_{m\to\infty} F_G(L_m) = \mathsf{E}\left[\lim_{m\to\infty} H^\star(L_m - A)\right] \tag{179}$$
$$= \mathsf{E}[H^\star(L - A)] \tag{180}$$
$$= F_G(L), \tag{181}$$

where (179) uses bounded convergence, (180) uses continuity of $H^\star$, and (181) recalls (26). The same argument and (176)–(178) give the limits zero and one for F_G. Its strict increase follows by averaging the pointwise strict inequality (174). Thus $F_G(L) = 1 - \varepsilon$ has a unique solution for every $\varepsilon \in (0, 1)$, proving Proposition 1.

APPENDIX B
PROOF OF COROLLARY 1

We first compare $F_G(L)$ with $\Phi(\sqrt{2}L)$ and invert the inequality to prove Corollary 1. We then evaluate the specialization in Section III-B1. Recall that A is independent of the Gaussian pair (J, B). From (23),

$$\begin{aligned}
\mathsf{Var}(J+B) &= \Sigma_{11} + 2\Sigma_{12} + \Sigma_{22} && (182)\\
&= \rho - \frac{1}{2}\left(\left(\rho - \frac{D}{q}\right)^2 + 2\frac{D}{q}\left(\rho - \frac{D}{q}\right) + \left(\frac{D}{q}\right)^2\right) && (183)\\
&= \rho - \frac{\rho^2}{2}, && (184)
\end{aligned}$$

where (182) expands the variance of a sum, (183) substitutes the covariance entries, and (184) collects the square. Consequently,

$$\begin{aligned}
\mathsf{Var}(A+J+B) &= \mathsf{Var}(A) + \mathsf{Var}(J+B) && (185)\\
&= \frac{(1-\rho)^2}{2} + \rho - \frac{\rho^2}{2} && (186)\\
&= \frac{1}{2}, && (187)
\end{aligned}$$

where (185) uses independence, (186) recalls the variance of A and (184), and (187) simplifies the expression.

For each $z \in \mathbb{R}$, Proposition 1 supplies a finite maximizing allocation $s^\star(z)$. The event $\{J \le z - s^\star(z), B \le s^\star(z)\}$ lies in $\{J + B \le z\}$ and excludes the nonempty open set $\{J > z - s^\star(z), B < z - J\}$. Since the Gaussian density is positive everywhere, $H^\star(z) < \mathbb{P}\{J + B \le z\}$. Averaging over A gives

$$\begin{aligned}
F_G(L) &= \mathsf{E}[H^\star(L - A)] && (188)\\
&< \mathsf{E}\big[\mathbb{P}\{J + B \le L - A \mid A\}\big] && (189)\\
&= \mathbb{P}\{A + J + B \le L\} && (190)\\
&= \Phi(\sqrt{2}L), && (191)
\end{aligned}$$

where (188) recalls (26), (189) uses the strict rectangle inclusion and independence of A and (J, B), (190) uses the tower property, and (191) uses the centered Gaussian distribution with variance (187). At the unique quantile $L_G(D, \varepsilon)$,

$$\begin{aligned}
1 - \varepsilon &= F_G(L_G(D, \varepsilon)) && (192)\\
&< \Phi\big(\sqrt{2}L_G(D, \varepsilon)\big), && (193)
\end{aligned}$$

where (192) follows from Proposition 1 and (193) applies (191). Therefore,

$$\begin{aligned}
L_{\mathrm{both}}(\varepsilon) &= \frac{1}{\sqrt{2}}\Phi^{-1}(1 - \varepsilon) && (194)\\
&< L_G(D, \varepsilon) && (195)\\
&= L^\star_{\mathrm{WZ}}(D, \varepsilon), && (196)
\end{aligned}$$

where (194) recalls (18), (195) uses strict monotonicity of Φ in (193), and (196) follows from Theorem 1.

This completes the proof of Corollary 1.

A. Evaluation for the Numerical Example

We derive (32) by first considering $D = S/2$. The definitions of S and ρ in (13) and (9) give $D/q = \rho/2$, so the two diagonal entries of Σ in (23) are equal. Hence

$$\begin{aligned}
\mathsf{Cov}(J + B, J - B) &= \mathsf{Var}(J) - \mathsf{Var}(B) && (197)\\
&= 0, && (198)
\end{aligned}$$

where (197) expands the covariance and (198) uses equality of the two variances. Joint Gaussianity then makes $J + B$ and $J - B$ independent.

Fix $z, s \in \mathbb{R}$ and a value $\xi \le z$ of $J + B$. The conditional rectangle probability satisfies

$$\begin{aligned}
\mathbb{P}\{J \le z - s, B \le s \mid J + B = \xi\} &= \mathbb{P}\{\xi - 2s \le J - B \le 2z - 2s - \xi\} && (199)\\
&= \mathbb{P}\{|(J - B) - (z - 2s)| \le z - \xi\} && (200)\\
&\le \mathbb{P}\{|J - B| \le z - \xi\}, && (201)
\end{aligned}$$

where (199) rewrites the two constraints and uses independence of the sum and difference, (200) identifies the interval's center and radius, and (201) uses symmetry and decrease of the centered Gaussian density away from zero. The choice $s = z/2$ attains the last bound for every ξ simultaneously; the rectangle is empty when $\xi > z$. Thus

$$H^\star(z) = \Phi_\Sigma(z/2, z/2) \tag{202}$$
$$= \mathbb{P}\{\max\{2J, 2B\} \le z\} \tag{203}$$
$$= \mathbb{P}\{J + B + |J - B| \le z\}, \tag{204}$$

where (202) follows by averaging the conditionally optimal allocation, (203) rewrites the rectangle, and (204) uses the identity $\max\{2J, 2B\} = J + B + |J - B|$.

The variance of the difference is

$$\mathsf{Var}(J - B) = \Sigma_{11} + \Sigma_{22} - 2\Sigma_{12} \tag{205}$$
$$= \rho\left(1 - \frac{\rho}{4}\right) + \frac{\rho^2}{4} \tag{206}$$
$$= \rho, \tag{207}$$

where (205) expands the variance and (206) substitutes $D/q = \rho/2$ into (23). Let G_c and G_a be independent standard Gaussian variables. Equations (187) and (207), together with independence of A and (J, B), identify the joint distribution of $(A + J + B, J - B)$. We obtain

$$F_G(L) = \mathsf{E}\big[\mathbb{P}\{J + B + |J - B| \le L - A \mid A\}\big] \tag{208}$$
$$= \mathbb{P}\{A + J + B + |J - B| \le L\} \tag{209}$$
$$= \mathbb{P}\left\{\frac{G_c}{\sqrt{2}} + \sqrt{\rho}\,|G_a| \le L\right\}, \tag{210}$$

where (208) combines (26) and (204), (209) uses the tower property, and (210) uses the independent centered Gaussian sum and difference with the variances just calculated.

Finally, let $p = q = 2$ and $D = 1/2$. Then $\rho = 1/2$ and $\mathsf{Var}(A) = 1/8$. Substitution into (23) gives

$$\Sigma = \begin{pmatrix} 7/32 & -1/32 \\ -1/32 & 7/32 \end{pmatrix}. \tag{211}$$

The covariance matrix of $(A + 2J, A + 2B)$ can now be evaluated as

$$\mathrm{Cov}\begin{pmatrix} A + 2J \\ A + 2B \end{pmatrix} = \mathsf{Var}(A)\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + 4\Sigma \tag{212}$$
$$= \frac{1}{8}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{8}\begin{pmatrix} 7 & -1 \\ -1 & 7 \end{pmatrix} \tag{213}$$
$$= I_2, \tag{214}$$

where (212) uses independence of A and (J, B), (213) uses (211) and $\mathsf{Var}(A) = 1/8$, and (214) adds the matrices. Both coordinates are centered, so their joint Gaussian distribution has independent standard Gaussian marginals. Consequently,

$$F_G(L) = \mathbb{P}\{A + 2J \le L, A + 2B \le L\} \tag{215}$$
$$= \mathbb{P}\{A + 2J \le L\}\mathbb{P}\{A + 2B \le L\} \tag{216}$$
$$= \Phi(L)^2, \tag{217}$$

where (215) averages (202) over A, (216) uses independence, and (217) uses the standard Gaussian marginals. Strict monotonicity of Φ then gives

$$L_G(1/2, \varepsilon) = \inf\{L \in \mathbb{R} : \Phi(L)^2 \ge 1 - \varepsilon\} \tag{218}$$
$$= \Phi^{-1}(\sqrt{1 - \varepsilon}), \tag{219}$$

where (218) uses (27) and (217), and (219) takes the positive square root and inverts the increasing CDF. Theorem 1 now gives (32).

For completeness, the folded-Gaussian representation also gives an integral for evaluating the second-order coding rate when $D = S/2$:

$$F_G(L) = \mathsf{E}\big[\Phi\big(\sqrt{2}(L - \sqrt{\rho}\,|G_a|)\big)\big] \tag{220}$$
$$= 2\int_0^\infty \varphi(g)\Phi\big(\sqrt{2}(L - \sqrt{\rho}\,g)\big)\,dg, \tag{221}$$

where (220) conditions (210) on G_a and uses independence, and (221) uses the density $2\varphi(g)$ of $|G_a|$ for $g \geq 0$. In the accompanying code, we evaluate (221) by adaptive quadrature and solve $F_G(L) = 1 - \varepsilon$ by bracketed root finding. Truncating the integral at $g = 9$ has error at most $2(1 - \Phi(9))$; the absolute quadrature tolerance is 10^{-11} and the root tolerance is 10^{-10}. The closed form (217) verifies the integral at $\rho = 1/2$. For $\varepsilon = 0.1$, the second-order coding rates in (32) and (18) are approximately 1.6322187896 and 0.9061938024 nats, respectively. Substituting them into (33) and (34) gives Fig. 2; the difference is approximately $0.7260249872/\sqrt{n}$ nats per source symbol.

APPENDIX C
PROOF OF LEMMA 1

Recall the Gaussian reference distribution $\bar{Q}$, residual variance T in (37), and shell $\mathcal{E}_n(x^n)$ in (57). Fix L, b, δ, p, q, D throughout this proof. All constants may depend on these parameters. We first lower bound the shell probability $\alpha_n(x^n)$ uniformly over $x^n \in \mathcal{C}_{n,b}$. We then establish the three limits in (66) and calculate the conditional means $\bar{d}$ and $\bar{g}$ for $u^n \in \mathcal{E}_n(x^n)$.

A. Probability of the Source-Dependent Shell

Define the conditional residual variance

$$\sigma_{\mathrm{a}}^2 := T\left(1 - \frac{T}{p}\right). \tag{222}$$

Positivity follows from $0 < T < p$. Gaussian conditioning under $\bar{Q}$ gives

$$X^n - U^n \mid X^n = x^n \sim \mathcal{N}\left(\frac{T}{p}x^n, \sigma_{\mathrm{a}}^2 I_n\right), \tag{223}$$

where (223) follows from the Gaussian conditional distribution (39) and independence across coordinates; the conditional residual mean is $(T/p)x^n$ and its covariance is $(T - T^2/p)I_n$.

Fix any $x^n \in \mathcal{C}_{n,b}$. Its normalized squared norm satisfies

$$\frac{\|x^n\|^2}{n} = p + \frac{2pA_n(x^n)}{(1-\rho)\sqrt{n}} \tag{224}$$

$$= p + O_b(n^{-1/2}), \tag{225}$$

where (224) rearranges (21), and (225) uses $|A_n(x^n)| \leq b$. For any $u^n \in \mathcal{E}_n(x^n)$, the normalized squared residual satisfies

$$\frac{\|x^n - u^n\|^2}{n} = \frac{r_n(x^n)}{n} + O(n^{-1}) \tag{226}$$

$$= T - \frac{s(x^n)}{\lambda(1 - D/q)^2\sqrt{n}} + O(n^{-1}) \tag{227}$$

$$= T + O_b(n^{-1/2}), \tag{228}$$

where (226) uses the fixed shell width, (227) substitutes (56), and (228) uses local boundedness of $s^\star$ in Proposition 1. These bounds are deterministic and uniform on the retained shell.

Choose orthonormal coordinates whose first axis is parallel to x^n. Let $G_1, \ldots, G_n$ be independent standard Gaussian variables. The conditional distribution in (223) permits the representation

$$\|x^n - U^n\|^2 = \left(\frac{T}{p}\|x^n\| + \sigma_{\mathrm{a}}G_1\right)^2 + \sigma_{\mathrm{a}}^2 \sum_{j=2}^{n} G_j^2, \tag{229}$$

where (229) follows from orthogonal invariance of the centered Gaussian residual. For $g \in \mathbb{R}$, define the interval

$$\mathcal{I}_n(x^n, g) := \left\{ t > 0 : \left| \left(\frac{T}{p}\|x^n\| + \sigma_{\mathrm{a}}g\right)^2 + \sigma_{\mathrm{a}}^2 t - r_n(x^n) \right| \leq \delta \right\}. \tag{230}$$

For $|g| \leq 1$, the center of $\mathcal{I}_n(x^n, g)$ is $n - 1 + O_b(\sqrt{n})$ by (225), (56), and $(T/p)^2 p + \sigma_{\mathrm{a}}^2 = T$. Hence, there is a constant $K_b > 0$ such that, for all sufficiently large n, every interval $\mathcal{I}_n(x^n, g)$ with $x^n \in \mathcal{C}_{n,b}$ and $|g| \leq 1$ lies in $\{t : |t - (n-1)| \leq K_b\sqrt{n}\} \subset (0, \infty)$. Each interval has length $2\delta/\sigma_{\mathrm{a}}^2$.

Let f_m denote the density of a chi-square variable with m degrees of freedom, and let Γ denote the gamma function. Uniformly for z in a fixed compact interval, its local density satisfies

$$f_m(m + z\sqrt{m}) = \frac{(m + z\sqrt{m})^{m/2-1} e^{-(m+z\sqrt{m})/2}}{2^{m/2}\Gamma(m/2)} \tag{231}$$

$$= \frac{\exp\{(m/2 - 1)\log(1 + z/\sqrt{m}) - z\sqrt{m}/2\}}{\sqrt{4\pi m}}\big(1 + O(m^{-1})\big) \tag{232}$$

$$= \frac{e^{-z^2/4}}{\sqrt{4\pi m}}\big(1 + O(m^{-1/2})\big), \tag{233}$$

where (231) is the chi-square density, (232) uses Stirling's formula for $\Gamma(m/2)$, and (233) expands $\log(1+z/\sqrt{m})$ through its quadratic term. Compactness of the z-range makes both remainders uniform.

Recall that φ is the standard Gaussian density. With a sufficiently small constant $c_b > 0$, we obtain

$$\alpha_n(x^n) = \int_{\mathbb{R}} \varphi(g) \int_{\mathcal{I}_n(x^n,g)} f_{n-1}(t)\,dt\,dg \tag{234}$$

$$\geq \int_{-1}^{1} \varphi(g) \int_{\mathcal{I}_n(x^n,g)} f_{n-1}(t)\,dt\,dg \tag{235}$$

$$\geq \mathbb{P}\{|G_1| \leq 1\} \frac{2\delta}{\sigma_{\mathrm{a}}^2} \inf_{|t-(n-1)|\leq K_b\sqrt{n}} f_{n-1}(t) \tag{236}$$

$$\geq c_b n^{-1/2}, \tag{237}$$

where (234) follows from (229) and independence of G_1 and $\sum_{j=2}^n G_j^2$, (235) restricts the nonnegative integral to $|g| \leq 1$, (236) uses the length and containment of $\mathcal{I}_n(x^n, g)$, and (237) applies (233) with $m = n-1$. The constant c_b is independent of $x^n \in \mathcal{C}_{n,b}$, proving (65).

B. Squared Norms, Inner Product, and Conditional Means

The source and residual energy limits follow from (225) and (228). For the cross term, (223) gives

$$(x^n)^\top (x^n - U^n) \mid X^n = x^n \sim \mathcal{N}\left(\frac{T}{p}\|x^n\|^2, \sigma_{\mathrm{a}}^2\|x^n\|^2\right), \tag{238}$$

where (238) applies a linear functional to the conditional Gaussian residual. Its normalized mean differs from T by $O_b(n^{-1/2})$, and its normalized variance is at most $2p\sigma_{\mathrm{a}}^2/n$ for all sufficiently large n, by (225). Consequently,

$$R_{U^n|X^n=x^n}\left\{\left|\tfrac{(x^n)^\top(x^n-U^n)}{n} - T\right| > n^{-1/4}\right\} = \frac{\bar{Q}\{|(x^n)^\top(x^n-U^n)/n - T| > n^{-1/4},\ U^n \in \mathcal{E}_n(x^n) \mid X^n = x^n\}}{\alpha_n(x^n)} \tag{239}$$

$$\leq \frac{\bar{Q}\{|(x^n)^\top(x^n-U^n)/n - T| > n^{-1/4} \mid X^n = x^n\}}{\alpha_n(x^n)} \tag{240}$$

$$\leq \frac{2\sqrt{n}}{c_b} \exp\left\{-\frac{\sqrt{n}}{16p\sigma_{\mathrm{a}}^2}\right\}, \tag{241}$$

where (239) substitutes (59), (240) removes $\{U^n \in \mathcal{E}_n(x^n)\}$ from the numerator, and (241) uses (237) and the Gaussian tail bound. In the last step, the normalized conditional mean differs from T by at most $n^{-1/4}/2$ for all sufficiently large n. Since the upper bound in (241) tends to zero uniformly over $x^n \in \mathcal{C}_{n,b}$, the inner-product limit follows. Together with the two squared-norm bounds above, this proves (66).

For the conditional distortion mean, we have uniformly on $\mathcal{E}_n(x^n)$,

$$\bar{d}(x^n, u^n) = \left(1 - \frac{D}{q}\right)^2 \|x^n - u^n\|^2 + \frac{nD^2}{q} \tag{242}$$

$$= \left(1 - \frac{D}{q}\right)^2 \left(nT - \frac{\sqrt{n}\,s(x^n)}{\lambda(1-D/q)^2} + O(1)\right) + \frac{nD^2}{q} \tag{243}$$

$$= n\left(\left(1 - \frac{D}{q}\right)^2 T + \frac{D^2}{q}\right) - \frac{\sqrt{n}}{\lambda} s(x^n) + O_b(1) \tag{244}$$

$$= nD - \frac{\sqrt{n}}{\lambda} s(x^n) + O_b(1), \tag{245}$$

where (242) uses the conditional distortion mean (48), (243) uses the shell definition (57) and the radius (56), (244) collects terms of equal order, and (245) uses the residual variance definition (37). Similarly, the information mean satisfies

$$\bar{g}(x^n, u^n) = nR_{\mathrm{WZ}}(D) + \sqrt{n}A_n(x^n) - \lambda\big(\bar{d}(x^n, u^n) - nD\big) \tag{246}$$

$$= nR_{\mathrm{WZ}}(D) + \sqrt{n}A_n(x^n) + \sqrt{n}s(x^n) + O_b(1) \tag{247}$$

$$= nR_{\mathrm{WZ}}(D) + \sqrt{n}\big(A_n(x^n) + s(x^n)\big) + O_b(1), \tag{248}$$

where (246) rearranges (53), (247) uses (245), and (248) collects the second-order terms. This proves (67) and (68), completing the proof of Lemma 1.

APPENDIX D
PROOF OF LEMMA 2

Recall J_n and B_n from (69) and (70). We first derive their joint Gaussian limit for fixed (x^n, u^n) satisfying the three limits in (66). We then calculate the limiting covariance and prove that the CDF approximation is uniform over $x^n \in \mathcal{C}_{n,b}$ and all thresholds.

A. Conditional Gaussian Limit

Fix a source-description pair (x^n, u^n) and derive the conditional Gaussian approximation over $Z^n \sim \mathcal{N}(0_n, qI_n)$. Recall the scores J_n and B_n in (69) and (70), and define the residual vector $w^n := x^n - u^n$. The distortion score satisfies

$$B_n = \frac{\lambda}{\sqrt{n}} \left(\left\| \left(1 - \frac{D}{q}\right) w^n - \frac{D}{q} Z^n \right\|^2 - \bar{d}(x^n, u^n) \right) \tag{249}$$

$$= \frac{\lambda}{\sqrt{n}} \left(-2\frac{D}{q}\left(1 - \frac{D}{q}\right)(w^n)^\top Z^n + \frac{D^2}{q^2}(\|Z^n\|^2 - nq) \right) \tag{250}$$

$$= -\frac{q - D}{q^2} \frac{(w^n)^\top Z^n}{\sqrt{n}} + \frac{D}{2q^2} \frac{\|Z^n\|^2 - nq}{\sqrt{n}}, \tag{251}$$

where (249) substitutes (38) into (70), (250) expands the squared norm and uses (48), and (251) uses the multiplier definition (20). The sum of the two scores satisfies

$$J_n + B_n = \frac{\|(1-\rho)x^n - \rho Z^n\|^2 - \mathsf{E}_{Z^n}[\|(1-\rho)x^n - \rho Z^n\|^2]}{2S\sqrt{n}} \tag{252}$$

$$= -\frac{\rho(1-\rho)}{S} \frac{(x^n)^\top Z^n}{\sqrt{n}} + \frac{\rho^2}{2S} \frac{\|Z^n\|^2 - nq}{\sqrt{n}} \tag{253}$$

$$= -\frac{1}{p+q} \frac{(x^n)^\top Z^n}{\sqrt{n}} + \frac{\rho}{2q} \frac{\|Z^n\|^2 - nq}{\sqrt{n}}, \tag{254}$$

where (252) sums and centers (44), which cancels the centered distortion, (253) expands the remaining centered squared norm, and (254) uses the definitions of S and ρ in (13) and (9). Thus the residual projection enters the information and distortion scores with opposite signs and cancels from their sum.

Consider any sequence of fixed pairs (x^n, u^n) whose normalized squared norms and inner product satisfy (66). The inner products with the Gaussian noise are jointly Gaussian, with

$$\frac{1}{\sqrt{n}} \begin{pmatrix} (x^n)^\top Z^n \\ (w^n)^\top Z^n \end{pmatrix} \sim \mathcal{N}\left(0_2, \frac{q}{n} \begin{pmatrix} \|x^n\|^2 & (x^n)^\top w^n \\ (x^n)^\top w^n & \|w^n\|^2 \end{pmatrix}\right), \tag{255}$$

where (255) calculates the covariance of the two inner products using $Z^n \sim \mathcal{N}(0_n, qI_n)$. Choose orthonormal coordinates so that x^n and w^n lie in the first two coordinate directions, adding any orthogonal direction if the two vectors are collinear. In these coordinates, both inner products depend only on Z_1, Z_2, and

$$\frac{\|Z^n\|^2 - nq}{\sqrt{n}} = \frac{Z_1^2 + Z_2^2 - 2q}{\sqrt{n}} + \frac{1}{\sqrt{n}} \sum_{i=3}^{n} (Z_i^2 - q). \tag{256}$$

The contribution of Z_1 and Z_2 to the centered squared norm is negligible because

$$\mathsf{E}_{Z^n}\left[\left(\frac{Z_1^2 + Z_2^2 - 2q}{\sqrt{n}}\right)^2\right] = \frac{\mathrm{Var}(Z_1^2) + \mathrm{Var}(Z_2^2)}{n} \tag{257}$$

$$= \frac{4q^2}{n}, \tag{258}$$

where (257) uses independence and centering, and (258) uses the fourth moment of a centered Gaussian. Thus, $n^{-1/2}\sum_{j=1}^{2}(Z_j^2 - q)$ tends to zero in mean square, with a bound independent of x^n, w^n. The sum over $j = 3, \ldots, n$ in (256) is independent of the two inner products and converges to $\mathcal{N}(0, 2q^2)$ by the central limit theorem. Indeed, the variables $Z_j^2 - q$ are independent with mean zero and variance $2q^2$. The covariance in (255) converges by (66). Consequently, (J_n, B_n) converges to a centered Gaussian pair, denoted temporarily by $(\bar{J}, \bar{B})$.

B. Covariance of the Limiting Pair

Odd Gaussian moments imply that each linear projection is uncorrelated with the centered quadratic term. Therefore,

$$\mathrm{Var}(\bar{B}) = \frac{(q-D)^2}{q^4} qT + \frac{D^2}{4q^4} 2q^2 \tag{259}$$

$$= \frac{D(q-D)}{q^2} + \frac{D^2}{2q^2} \tag{260}$$

$$= \frac{D}{q} - \frac{1}{2}\left(\frac{D}{q}\right)^2, \tag{261}$$

where (259) uses (251), (255), and the limiting quadratic variance, (260) substitutes the definition of T in (37), and (261) collects the powers of D/q. Similarly,

$$\mathrm{Var}(\bar{J}+\bar{B}) = \frac{qp}{(p+q)^2} + \frac{\rho^2}{2} \tag{262}$$

$$= \rho(1-\rho) + \frac{\rho^2}{2} \tag{263}$$

$$= \rho - \frac{\rho^2}{2}, \tag{264}$$

where (262) uses (254), (263) substitutes the definition of ρ in (9), and (264) simplifies the resulting expression. The covariance with the distortion score is

$$\mathrm{Cov}(\bar{J}+\bar{B}, \bar{B}) = \frac{q-D}{q^2(p+q)} qT + \frac{\rho D}{4q^3} 2q^2 \tag{265}$$

$$= \frac{D}{p+q} + \frac{\rho D}{2q} \tag{266}$$

$$= \frac{D}{q}\left(1 - \frac{\rho}{2}\right), \tag{267}$$

where (265) follows from (251), (254), and the inner-product limit $(x^n)^\top w^n/n \to T$, (266) uses the definition of T in (37), and (267) uses the definition of ρ in (9).

The off-diagonal covariance entry now follows from

$$\mathrm{Cov}(\bar{J}, \bar{B}) = \mathrm{Cov}(\bar{J}+\bar{B}, \bar{B}) - \mathrm{Var}(\bar{B}) \tag{268}$$

$$= \frac{D}{q}\left(1 - \frac{\rho}{2}\right) - \frac{D}{q} + \frac{D^2}{2q^2} \tag{269}$$

$$= -\frac{D}{2q}\left(\rho - \frac{D}{q}\right), \tag{270}$$

where (268) uses bilinearity of covariance, (269) combines (261) and (267), and (270) factors the difference. Finally,

$$\mathrm{Var}(\bar{J}) = \mathrm{Var}(\bar{J}+\bar{B}) + \mathrm{Var}(\bar{B}) - 2\,\mathrm{Cov}(\bar{J}+\bar{B}, \bar{B}) \tag{271}$$

$$= \rho - \frac{\rho^2}{2} + \frac{D}{q} - \frac{D^2}{2q^2} - \frac{2D}{q}\left(1 - \frac{\rho}{2}\right) \tag{272}$$

$$= \left(\rho - \frac{D}{q}\right)\left(1 - \frac{1}{2}\left(\rho - \frac{D}{q}\right)\right), \tag{273}$$

where (271) uses $\bar{J} = (\bar{J}+\bar{B}) - \bar{B}$, (272) substitutes the three covariance calculations, and (273) factors the resulting quadratic polynomial. Equations (261), (270), and (273) equal the entries of Σ in (23). Thus $(\bar{J}, \bar{B})$ has the same distribution as the pair (J, B) in the main result.

C. Uniformity and Averaging Over the Shell

For each $x^n \in \mathcal{C}_{n,b}$, define the descriptions whose normalized inner product is close to T:

$$\mathcal{V}_{n,b}(x^n) := \left\{ u^n \in \mathcal{E}_n(x^n) : \left| \frac{(x^n)^\top (x^n - u^n)}{n} - T \right| \le n^{-1/4} \right\}. \tag{274}$$

For all sufficiently large n, $R_{U^n|X^n=x^n}(\mathcal{V}_{n,b}(x^n)) > 0$ by (241). For a fixed pair (x^n, u^n), let F_{n,x^n,u^n} denote the CDF of (J_n, B_n) over Z^n. Define the largest CDF approximation error over the sets $\mathcal{V}_{n,b}(x^n)$ by

$$\eta_{n,b} := \sup_{x^n \in \mathcal{C}_{n,b}} \sup_{u^n \in \mathcal{V}_{n,b}(x^n)} \sup_{r,s \in \mathbb{R}} |F_{n,x^n,u^n}(r,s) - \Phi_\Sigma(r,s)|. \tag{275}$$

Every sequence with $x^n \in \mathcal{C}_{n,b}$ and $u^n \in \mathcal{V}_{n,b}(x^n)$ satisfies the three limits in (66). The conditional second moments of (J_n, B_n) are uniformly bounded by (251)–(255). Since Σ is positive definite, the finite-grid argument in [7, Appendix E-B, proof of Proposition 3] applies to the Gaussian limit along every such sequence and gives $\eta_{n,b} \to 0$. For any $x^n \in \mathcal{C}_{n,b}$ and $(r, s) \in \mathbb{R}^2$, we obtain

$$\begin{aligned}
\left|\widehat{P}\{J_n \le r, B_n \le s \mid X^n = x^n\} - \Phi_\Sigma(r,s)\right| &= \left|\int \big(F_{n,x^n,u^n}(r,s) - \Phi_\Sigma(r,s)\big) R_{U^n|X^n=x^n}(du^n)\right| && (276)\\
&\le \int \left|F_{n,x^n,u^n}(r,s) - \Phi_\Sigma(r,s)\right| R_{U^n|X^n=x^n}(du^n) && (277)\\
&\le \eta_{n,b} R_{U^n|X^n=x^n}(\mathcal{V}_{n,b}(x^n)) + R_{U^n|X^n=x^n}(\mathcal{V}_{n,b}(x^n)^c) && (278)\\
&\le \eta_{n,b} + \frac{2\sqrt{n}}{c_b} \exp\left\{-\frac{\sqrt{n}}{16p\sigma_{\mathrm{a}}^2}\right\}, && (279)
\end{aligned}$$

where (276) conditions on U^n, (277) uses the triangle inequality, (278) splits the integral over $\mathcal{V}_{n,b}(x^n)$ and its complement and bounds each CDF difference by one, and (279) applies (241). The upper bound in (279) tends to zero uniformly over $x^n \in \mathcal{C}_{n,b}$ and $(r, s) \in \mathbb{R}^2$, proving (71).

Finally, the conditional limit of (J_n, B_n) does not depend on $x^n \in \mathcal{C}_{n,b}$. Consider a product of bounded continuous functions, one of $A_n(X^n)$ and the other of (J_n, B_n). Conditioning on X^n and using (279) separates the limiting expectation into the product of the two expectations on each fixed cutoff. The omitted Gaussian source probability tends to zero as b increases. Hence the limiting pair (J, B) can be taken independent of A, completing the proof of Lemma 2.

APPENDIX E
PROOF OF LEMMA 3

Recall the indicator F_n in (93), the distribution $\tilde{P}$ conditioned on $F_n = 1$ in (94), and the sampled description V^{pre} with permutation Π. We first use a list of candidate reconstructions and V^{pre} to obtain expected distortion $D + o(1)$ under $\tilde{P}$. We then bound $K(t_0)$ and vary the Gaussian observation strength t to obtain the posterior covariance bound. Throughout this proof, $h_n = \sqrt{8p \log n}$ is the clipping level defined above (100).

A. Reconstruction From the Sampled Description

Fix a source vector x^n satisfying $F_n(x^n) = 1$. Define the observations for which the original decoder reconstructs x^n within distortion D:

$$\mathcal{A}_{x^n} := \{y^n : \|x^n - \phi(f(x^n), y^n)\|^2 \le nD\}. \tag{280}$$

By the flag definition, its probability under $\mathcal{N}(x^n, qI_n)$ is at least n^{-1}. Fix $\theta \in (0, 1)$, let G_0^n, G_1^n be jointly standard Gaussian vectors with cross-covariance θI_n, and let Γ_n denote standard Gaussian measure. For Borel sets $\mathcal{E}, \mathcal{A} \subseteq \mathbb{R}^n$,

$$P(G_0^n \in \mathcal{E}, G_1^n \in \mathcal{A}) \ge [\Gamma_n(\mathcal{E})\Gamma_n(\mathcal{A})]^{1/(1-\theta)}, \tag{281}$$

where (281) follows from the reverse Hölder inequality in [13, Theorem 1(ii)]. Its covariance hypothesis holds because the matrix $\left(\begin{smallmatrix} I_n & \theta I_n \\ \theta I_n & I_n \end{smallmatrix}\right) - (1-\theta) I_{2n}$ is positive semidefinite.

Let $\bar{G}^n$ be an independent standard Gaussian vector, so that G_1^n can be represented as $\theta G_0^n + \sqrt{1-\theta^2}\bar{G}^n$. Given $b > 0$ and a Borel set $\mathcal{A}$ with $\Gamma_n(\mathcal{A}) > 0$, define the values of G_0^n for which the conditional probability of $G_1^n \in \mathcal{A}$ is smaller than e^{-b}:

$$\mathcal{E}_b := \{g^n : P(\theta g^n + \sqrt{1-\theta^2}\bar{G}^n \in \mathcal{A}) < e^{-b}\}. \tag{282}$$

Then

$$\begin{aligned}
[\Gamma_n(\mathcal{E}_b)\Gamma_n(\mathcal{A})]^{1/(1-\theta)} &\le P(G_0^n \in \mathcal{E}_b, G_1^n \in \mathcal{A}) && (283)\\
&= \int_{\mathcal{E}_b} P(\theta g^n + \sqrt{1-\theta^2}\bar{G}^n \in \mathcal{A})\, \Gamma_n(dg^n) && (284)\\
&\le e^{-b}\Gamma_n(\mathcal{E}_b), && (285)
\end{aligned}$$

where (283) uses (281), (284) conditions on G_0^n, and (285) uses (282). Rearranging gives

$$\Gamma_n(\mathcal{E}_b) \le \Gamma_n(\mathcal{A})^{-1/\theta} e^{-b(1-\theta)/\theta}. \tag{286}$$

The case $\Gamma_n(\mathcal{E}_b) = 0$ satisfies the same bound directly.

Choose $\delta_n := \sqrt{(\log n)/n}$, $\theta_n := 1 - \delta_n$, and $b_n^{\mathrm{list}} := (1 + 3\theta_n) \log n/\delta_n$. For all sufficiently large n, $\theta_n \in (0, 1)$. Conditional on $X^n = x^n$, the ideal perturbation

$$Y^{n,*} := \theta_n Y^n + \delta_n x^n + \sqrt{q(1-\theta_n^2)}\bar{G}^n \tag{287}$$

has distribution $\mathcal{N}(x^n, qI_n)$, because its mean is x^n and its noise covariance is qI_n. Apply (286) to the standardized set $(\mathcal{A}_{x^n} - x^n)/\sqrt{q}$. We obtain

$$P\big(P(Y^{n,*} \in \mathcal{A}_{x^n} \mid X^n = x^n, Y^n) < e^{-b_n^{\mathrm{list}}} \,\big|\, X^n = x^n\big) \le n^{1/\theta_n} \exp\{-b_n^{\mathrm{list}} \delta_n/\theta_n\} \tag{288}$$

$$= n^{-3}, \tag{289}$$

where (288) uses $P(Y^n \in \mathcal{A}_{x^n} \mid X^n = x^n) \ge n^{-1}$ and (289) substitutes b_n^{list}. Outside the event bounded in (289), the conditional probability that $Y^{n,*} \in \mathcal{A}_{x^n}$ is at least $e^{-b_n^{\mathrm{list}}}$.

For each $j \in \mathbb{N}$, the decoder generates a candidate using Y^n and independent Gaussian randomness: $Y^{n,(j)} := \theta_n Y^n + \sqrt{q(1-\theta_n^2)} G_j^n$, where the G_j^n are independent standard Gaussian vectors, independent of the original variables and Π. These candidates do not use x^n. To compare their success probabilities with the preceding source-dependent perturbation, let P_* and P_0 be Gaussian distributions with common covariance $\sigma^2 I_n$ and mean difference v_0^n. Cauchy–Schwarz gives

$$P_*(\mathcal{A}) = \mathsf{E}_{P_0}\left[\mathbb{1}\{\mathcal{A}\}\frac{dP_*}{dP_0}\right] \tag{290}$$

$$\le \sqrt{P_0(\mathcal{A})}\left(\mathsf{E}_{P_0}\left[\left(\frac{dP_*}{dP_0}\right)^2\right]\right)^{1/2} \tag{291}$$

$$= \sqrt{P_0(\mathcal{A})}\exp\{\|v_0^n\|^2/(2\sigma^2)\}, \tag{292}$$

where (290) changes measure, (291) applies Cauchy–Schwarz, and (292) evaluates the Gaussian exponential moment. Define

$$c_n := 2b_n^{\mathrm{list}} + \frac{2pn\delta_n}{q(1+\theta_n)}. \tag{293}$$

Since $\delta_n = \sqrt{(\log n)/n}$, we have $c_n = O(\sqrt{n \log n})$. Outside the observation event in (289), each candidate satisfies

$$P(Y^{n,(j)} \in \mathcal{A}_{x^n} \mid X^n = x^n, Y^n) \ge P(Y^{n,*} \in \mathcal{A}_{x^n} \mid X^n = x^n, Y^n)^2 \exp\left\{-\frac{\delta_n^2 \|x^n\|^2}{q(1-\theta_n^2)}\right\} \tag{294}$$

$$\ge \exp\left\{-2b_n^{\mathrm{list}} - \frac{2pn\delta_n}{q(1+\theta_n)}\right\} \tag{295}$$

$$= e^{-c_n}, \tag{296}$$

where (294) rearranges (292), (295) uses $\|x^n\|^2 \le 2pn$ and $1 - \theta_n = \delta_n$, and (296) uses (293).

Use the deterministic list size $N_n := \lceil ne^{c_n} \rceil$, and define the list-failure event by

$$\mathcal{E}_n^{\mathrm{list}} := \bigcap_{j \in [N_n]} \{Y^{n,(j)} \notin \mathcal{A}_{X^n}\}. \tag{297}$$

For every retained x^n,

$$P(\mathcal{E}_n^{\mathrm{list}} \mid X^n = x^n) \le n^{-3} + (1 - e^{-c_n})^{N_n} \tag{298}$$

$$\le n^{-3} + \exp\{-N_n e^{-c_n}\} \tag{299}$$

$$\le n^{-3} + e^{-n}, \tag{300}$$

where (298) uses (289), (296), and conditional independence of the seeds, (299) uses $1 - a \le e^{-a}$, and (300) uses the choice of N_n. The list size depends only on n, p, q.

Clip each reconstruction $\phi(f(X^n), Y^{n,(j)})$ coordinatewise to $[-h_n, h_n]$, obtaining $\hat{X}^{n,(j)}$. On the flag set, clipping cannot increase its error. Define the full-block loss by

$$\mathsf{L}_j := \frac{1}{n}\|X^n - \hat{X}^{n,(j)}\|^2 \tag{301}$$

and its sampled version by

$$\widehat{\mathsf{L}}_j := \frac{1}{k}\sum_{i=1}^{k}(X_{\Pi(i)} - \hat{X}^{(j)}_{\Pi(i)})^2. \tag{302}$$

Let $\mathcal{E}_n^{\mathrm{sam}} := \{\max_{j\in[N_n]} |\widehat{\mathsf{L}}_j - \mathsf{L}_j| > n^{-1/16}\}$. Conditional on the source, observation, and list seeds, all coordinate losses lie in $[0, 4h_n^2]$, and the list is independent of Π. For some constant $c_{\mathrm{s}} > 0$ and all sufficiently large n,

$$\tilde{P}(\mathcal{E}_n^{\mathrm{sam}}) \le \sum_{j=1}^{N_n} \tilde{P}(|\widehat{\mathsf{L}}_j - \mathsf{L}_j| > n^{-1/16}) \tag{303}$$

$$\le 2N_n \exp\left\{-\frac{kn^{-1/8}}{8h_n^4}\right\} \tag{304}$$

$$\le \exp\{-c_{\mathrm{s}} n^{5/8}/(\log n)^2\}, \tag{305}$$

where (303) is the union bound, (304) follows from the bounded-variable inequality and sampling comparison in [14, Theorems 2 and 4], and (305) uses $k = \lceil n^{3/4} \rceil$, $h_n^2 = 8p \log n$, and $\log N_n = O(\sqrt{n \log n})$. The magnitude of the negative exponent in (304) grows faster than $\log N_n$, so the union over the list still has vanishing probability.

Let $\check{\mathsf{L}}_j$ be the sample loss computed from the quantized prefix, and let $\hat{j}$ minimize $\check{\mathsf{L}}_j$, with the smallest index used for a tie. Define $\hat{X}^{n,\mathrm{rep}} := \hat{X}^{n,(\hat{j})}$. Replacing a sampled source coordinate by a representative in its quantization interval changes the squared loss by at most $4h_n\Delta_n + \Delta_n^2$. On $(\mathcal{E}_n^{\mathrm{list}} \cup \mathcal{E}_n^{\mathrm{sam}})^c$, choose an index $j^\star$ for which $\mathsf{L}_{j^\star} \le D$. Then

$$\frac{1}{n}\|X^n - \hat{X}^{n,\mathrm{rep}}\|^2 = \mathsf{L}_{\hat{j}} \tag{306}$$

$$\le \widehat{\mathsf{L}}_{\hat{j}} + n^{-1/16} \tag{307}$$

$$\le \check{\mathsf{L}}_{\hat{j}} + 4h_n\Delta_n + \Delta_n^2 + n^{-1/16} \tag{308}$$

$$\le \check{\mathsf{L}}_{j^\star} + 4h_n\Delta_n + \Delta_n^2 + n^{-1/16} \tag{309}$$

$$\le \mathsf{L}_{j^\star} + 8h_n\Delta_n + 2\Delta_n^2 + 2n^{-1/16} \tag{310}$$

$$\le D + 2n^{-1/16} + 8h_n\Delta_n + 2\Delta_n^2, \tag{311}$$

where (306) is the decoder definition, (307) uses $(\mathcal{E}_n^{\mathrm{sam}})^c$, (308) uses the quantization error bound, (309) uses the minimizing choice of $\hat{j}$, (310) applies the same quantization and sampling bounds to $j^\star$, and (311) uses $\mathsf{L}_{j^\star} \le D$. On $\mathcal{E}_n^{\mathrm{list}} \cup \mathcal{E}_n^{\mathrm{sam}}$, the loss is at most $4h_n^2$ because the source and reconstruction coordinates lie in $[-h_n, h_n]$. Therefore,

$$\mathsf{E}_{\tilde{P}}[\|X^n - \hat{X}^{n,\mathrm{rep}}\|^2] \le nD + 2n^{15/16} + 8nh_n\Delta_n + 2n\Delta_n^2 + 4nh_n^2\tilde{P}(\mathcal{E}_n^{\mathrm{list}} \cup \mathcal{E}_n^{\mathrm{sam}}) \tag{312}$$

$$\le nD + 2n^{15/16} + 8nh_n\Delta_n + 2n\Delta_n^2 + 4nh_n^2\big(n^{-3} + e^{-n} + e^{-c_{\mathrm{s}} n^{5/8}/(\log n)^2}\big) \tag{313}$$

$$= nD + O(n^{15/16}), \tag{314}$$

where (312) splits the expectation according to $\mathcal{E}_n^{\mathrm{list}} \cup \mathcal{E}_n^{\mathrm{sam}}$, (313) uses (300) uniformly over $F_n(x^n) = 1$ and (305), and (314) substitutes $\Delta_n = n^{-2}$ and $h_n = O(\sqrt{\log n})$. We have thus constructed the reconstruction needed to bound $K(t_0)$.

B. Posterior Relative Entropy

We next convert (314) to a relative entropy bound. Write $\tilde{p}_{X^n|Y^n}$ and $p_{X^n|Y^n}$ for the conditional source densities under $\tilde{P}$ and P, respectively. On the retained support, source-only conditioning gives

$$\tilde{p}_{X^n|Y^n}(x^n \mid y^n) = \frac{p_{X^n|Y^n}(x^n \mid y^n)F_n(x^n)}{P(F_n = 1 \mid Y^n = y^n)}. \tag{315}$$

Also, data processing gives $\mathrm{KL}(\tilde{P}_{Y^n}\|P_{Y^n}) \le \log(1/\beta_n)$, and Cauchy–Schwarz gives $|\mathsf{E}_{\tilde{P}}[\|X^n\|^2] - np| \le \sqrt{2np^2/\beta_n}$. Consequently,

$$h_{\tilde{P}}(X^n \mid Y^n) = \mathsf{E}_{\tilde{P}}[-\log p_{X^n|Y^n}(X^n \mid Y^n)] + \mathsf{E}_{\tilde{P}}[\log P(F_n = 1 \mid Y^n)] \tag{316}$$

$$= \mathsf{E}_{\tilde{P}}[-\log p_{X^n|Y^n}(X^n \mid Y^n)] + \log\beta_n + \mathrm{KL}(\tilde{P}_{Y^n}\|P_{Y^n}) \tag{317}$$

$$= \frac{n}{2}\log(2\pi S) + \frac{(1-\rho)^2\mathsf{E}_{\tilde{P}}[\|X^n\|^2] + n\rho^2 q}{2S} + \log\beta_n + \mathrm{KL}(\tilde{P}_{Y^n}\|P_{Y^n}) \tag{318}$$

$$= \frac{n}{2}\log(2\pi eS) + O(\sqrt{n}), \tag{319}$$

where (316) uses (315), (317) uses $d\tilde{P}_{Y^n}/dP_{Y^n} = P(F_n = 1 \mid Y^n)/\beta_n$, (318) evaluates the Gaussian conditional density and uses (99), and (319) uses β_n bounded away from zero, the preceding energy bound, and $(1-\rho)^2 p + \rho^2 q = S$.

Since W is deterministic given (X^n, Π) and Π is independent of (X^n, Y^n),

$$h_{\tilde{P}}(X^n \mid W, Y^n, \Pi) = h_{\tilde{P}}(X^n \mid Y^n) - H_{\tilde{P}}(W \mid Y^n, \Pi) \tag{320}$$

$$\ge \frac{n}{2}\log(2\pi eS) - \log(2M_n) - O(k\log n + \sqrt{n}) \tag{321}$$

$$\ge \frac{n}{2}\log(2\pi eD) - O(k\log n + \sqrt{n}), \tag{322}$$

where (320) is the conditional entropy chain rule, (321) uses (319) and (101), and (322) uses (90) and (15).

Let $m_0^n := \mathsf{E}_{\tilde{P}}[X^n \mid W, Y^n, \Pi]$, and recall $K(t)$ from (102). Gaussian cross entropy and conditional-mean orthogonality give

$$K(t_0) + \frac{\mathsf{E}_{\tilde{P}}[\|m_0^n - \hat{X}^{n,\mathrm{rep}}\|^2]}{2D} = \frac{n}{2}\log(2\pi D) - h_{\tilde{P}}(X^n \mid W, Y^n, \Pi) + \frac{\mathsf{E}_{\tilde{P}}[\|X^n - m_0^n\|^2] + \mathsf{E}_{\tilde{P}}[\|m_0^n - \hat{X}^{n,\mathrm{rep}}\|^2]}{2D} \tag{323}$$

$$= \frac{n}{2}\log(2\pi eD) - h_{\tilde{P}}(X^n \mid W, Y^n, \Pi) + \frac{\mathsf{E}_{\tilde{P}}[\|X^n - \hat{X}^{n,\mathrm{rep}}\|^2] - nD}{2D} \tag{324}$$

$$\le O(k\log n + \sqrt{n} + n^{15/16}) \tag{325}$$

$$= O(n^{15/16}), \tag{326}$$

where (323) expands the relative entropy, (324) uses conditional-mean orthogonality, (325) uses (322) and (314), and (326) uses $k = \lceil n^{3/4} \rceil$. The private list randomness is independent of X^n conditional on (W, Y^n, Π), so orthogonality remains valid after averaging that randomness. Both terms on the left are nonnegative, proving the bound on $K(t_0)$.

C. Choosing the Gaussian Observation Strength

Define $\upsilon(t) := \mathsf{E}_{\tilde{P}}[\operatorname{tr} V_t]$. The conditional vector I–MMSE identity yields

$$\frac{d}{dt} h_{\tilde{P}}(X^n \mid W, O_t^n, \Pi) = -\frac{1}{2}\upsilon(t), \tag{327}$$

where (327) follows by applying [11, Theorem 2] to each conditional input distribution given (W, Π) and averaging. The Gaussian-channel Hessian identity similarly gives

$$\upsilon'(t) = -\mathsf{E}_{\tilde{P}}[\operatorname{tr}(V_t^2)], \tag{328}$$

where (328) follows from [12, Corollary 1, Eq. (45)]. For fixed n, all localized conditional inputs have common bounded support, which justifies differentiation and averaging. Expanding the Gaussian cross entropy in (102), we obtain

$$K'(t) = \frac{nD_t'}{2D_t} + \frac{\upsilon'(t)}{2D_t} - \frac{\upsilon(t)D_t'}{2D_t^2} - \frac{d}{dt} h_{\tilde{P}}(X^n \mid W, O_t^n, \Pi) \tag{329}$$

$$= -\frac{nD_t}{2} + \upsilon(t) - \frac{\mathsf{E}_{\tilde{P}}[\operatorname{tr}(V_t^2)]}{2D_t} \tag{330}$$

$$= -\frac{1}{2D_t}\mathsf{E}_{\tilde{P}}[\operatorname{tr}(V_t^2) - 2D_t \operatorname{tr} V_t + nD_t^2] \tag{331}$$

$$= -\frac{1}{2D_t}\mathsf{E}_{\tilde{P}}[\|V_t - D_t I_n\|_{\mathrm{F}}^2], \tag{332}$$

where (329) differentiates the cross-entropy formula, (330) uses (327), (328), and $D_t' = -D_t^2$, (331) uses the definition of $\upsilon(t)$, and (332) expands the Frobenius norm. Thus K is nonnegative and nonincreasing. Since $D_t \le D$ for $t \ge t_0$,

$$\int_{t_0}^{t_0+n^{-1/32}} \mathsf{E}_{\tilde{P}}[\|V_t - D_t I_n\|_{\mathrm{F}}^2]\, dt = -2\int_{t_0}^{t_0+n^{-1/32}} D_t K'(t)\, dt \tag{333}$$

$$\le 2D\big(K(t_0) - K(t_0 + n^{-1/32})\big) \tag{334}$$

$$\le 2DK(t_0) \tag{335}$$

$$= O(n^{15/16}), \tag{336}$$

where (333) uses (332), (334) uses $D_t \le D$ and $K'(t) \le 0$, (335) uses $K \ge 0$, and (336) uses (326).

Averaging the covariance bound over the deterministic interval $[t_0, t_0 + n^{-1/32}]$ gives a deterministic $\nu_n \in (0, n^{-1/32}]$ such that

$$\mathsf{E}_{\tilde{P}}[\|V_{t_0+\nu_n} - D_{t_0+\nu_n} I_n\|_{\mathrm{F}}^2] \le 2n^{1/32}\int_{t_0}^{t_0+n^{-1/32}} \mathsf{E}_{\tilde{P}}[\|V_t - D_t I_n\|_{\mathrm{F}}^2]\, dt \tag{337}$$

$$= O(n^{31/32}) \tag{338}$$

$$= o(n), \tag{339}$$

where (337) follows from averaging, (338) uses (336), and (339) follows since $31/32 < 1$. Moreover,

$$K(t_0 + \nu_n) \le K(t_0) \tag{340}$$

$$= O(n^{15/16}) \tag{341}$$

$$= o(n), \tag{342}$$

where (340) uses monotonicity, (341) uses (326), and (342) follows since $15/16 < 1$. This completes the proof of Lemma 3.

APPENDIX F
PROOF OF LEMMA 4

Recall the added observation $A^{n,\mathrm{ext}}$, combined observation $\tilde{Y}^n$, variances $(\tilde{q}, \tilde{S}, \tilde{D})$, and likelihood $\tilde{H}$ from Sections V-A and V-B. We first calculate the change in message likelihood caused by $A^{n,\mathrm{ext}}$. We then evaluate the cost of V^{pre} and derive the upper and lower bounds in Lemma 4, retaining the same term b_n^{pre} in both. We use φ_{vI_d} as defined before (116), and write φ_v when $d = 1$.

A. Effect of the Added Observation

Define the Gaussian vector

$$V^{n,\mathrm{anc}} := \frac{A^{n,\mathrm{ext}} - \sqrt{\nu} Y^n}{\sqrt{1+\nu q}} \tag{343}$$

$$= \frac{G^{n,\mathrm{ext}} - \sqrt{\nu} Z^n}{\sqrt{1+\nu q}}, \tag{344}$$

where (343) defines the vector and (344) uses the observation model (1) and the definition of $A^{n,\mathrm{ext}}$. Its covariance is I_n, and its cross-covariance with $\tilde{Z}^n$ in (107) is zero. Joint Gaussianity and independence of the noises from (X^n, Π) imply that $V^{n,\mathrm{anc}}$ is independent of $(X^n, W, \tilde{Y}^n, \Pi)$. Inverting the linear transformation gives

$$Y^n = \tilde{Y}^n - \frac{q\sqrt{\nu}}{\sqrt{1+\nu q}} V^{n,\mathrm{anc}}, \tag{345}$$

where (345) uses (106), (110), and (344). Thus the original decoder can be evaluated from $(\tilde{Y}^n, V^{n,\mathrm{anc}})$.

Clip the original reconstruction $\phi(M, Y^n)$ coordinatewise to $[-h_n, h_n]$, with $h_n = \sqrt{8p \log n}$ as above (100), and denote the result by $\bar{X}^n$. Since $|X_i| \le h_n$ for every $i \in [n]$ on $\{F_n = 1\}$, clipping cannot increase the distortion there. Define $e^{n,\mathrm{c}} := X^n - \bar{X}^n$ and the reconstruction using the added observation by

$$\hat{X}^{n,\mathrm{enh}} := \frac{\bar{X}^n + D\sqrt{\nu} A^{n,\mathrm{ext}}}{1+\nu D}. \tag{346}$$

The reconstruction $\hat{X}^{n,\mathrm{enh}}$ uses neither V^{pre} nor Π. Its squared error satisfies

$$\|X^n - \hat{X}^{n,\mathrm{enh}}\|^2 - n\tilde{D} = \frac{\|e^{n,\mathrm{c}} - D\sqrt{\nu} G^{n,\mathrm{ext}}\|^2}{(1+\nu D)^2} - \frac{nD}{1+\nu D} \tag{347}$$

$$= \frac{\|e^{n,\mathrm{c}}\|^2 - nD - 2D\sqrt{\nu}\,(e^{n,\mathrm{c}})^\top G^{n,\mathrm{ext}} + D^2\nu(\|G^{n,\mathrm{ext}}\|^2 - n)}{(1+\nu D)^2}, \tag{348}$$

where (347) uses (346) and $\tilde{D} = D/(1+\nu D)$, and (348) expands the square. On $\mathcal{S}_n \cap \{F_n = 1\}$, the clipped error satisfies $\|e^{n,\mathrm{c}}\|^2 \le nD$. Conditional on (X^n, Y^n), the two centered noise terms in (348) are uncorrelated, with total variance $4D^2\nu\|e^{n,\mathrm{c}}\|^2 + 2nD^4\nu^2 = O(n\nu)$ on $\mathcal{S}_n \cap \{F_n = 1\}$. Thus, for every $\eta > 0$,

$$P\{\mathcal{S}_n, F_n = 1,\ \|X^n - \hat{X}^{n,\mathrm{enh}}\|^2 > n\tilde{D} + \eta\sqrt{n}\} \longrightarrow 0, \tag{349}$$

where (349) follows from conditional Chebyshev and $\nu \to 0$.

Let ϖ and κ be probability densities with respect to a common measure, and let P_ϖ denote the distribution with density ϖ. For every $\gamma > 0$,

$$P_\varpi\left(\log \frac{\varpi}{\kappa} < -\gamma\right) \le e^{-\gamma}, \tag{350}$$

where (350) is Markov's inequality applied to κ/ϖ, whose expectation under P_ϖ is at most one. The same bound applies to conditional densities and mass functions.

Write $p_P(\cdot \mid \cdot)$ for conditional densities under the original distribution P. Since $A^{n,\mathrm{ext}} \mid X^n = x^n \sim \mathcal{N}(\sqrt{\nu}x^n, I_n)$, the density $p_P(a^n \mid M_0, Y^n)$ is obtained by averaging the Gaussian channel density over $P_{X^n|M_0,Y^n}$. The latter distribution is $\mathcal{N}(\rho Y^n, SI_n)$ restricted and normalized on the source cell of M_0. Define the reference density

$$\kappa(a^n \mid M_0, Y^n) := \varphi_{(1+\nu D)I_n}(a^n - \sqrt{\nu}\bar{X}^n). \tag{351}$$

Here $\bar{X}^n$ is determined by (M_0, Y^n). Without the message, $A^{n,\mathrm{ext}} \mid Y^n \sim \mathcal{N}(\sqrt{\nu}\rho Y^n, (1+\nu S)I_n)$. Put $e_0^n := X^n - \rho Y^n$, and define

$$\mathsf{N}_n := \sqrt{\nu}\left(\frac{e_0^n}{1+\nu S} - \frac{e^{n,\mathrm{c}}}{1+\nu D}\right)^\top G^{n,\mathrm{ext}} + \frac{1}{2}\left(\frac{1}{1+\nu S} - \frac{1}{1+\nu D}\right)(\|G^{n,\mathrm{ext}}\|^2 - n). \tag{352}$$

For some constant C_1 depending only on (p, q, D), on $\mathcal{S}_n \cap \{F_n = 1\}$ we have

$$\mathrm{Var}_P(\mathsf{N}_n \mid X^n, Y^n) = \nu \left\| \frac{e_0^n}{1+\nu S} - \frac{e^{n,\mathrm{c}}}{1+\nu D} \right\|^2 + \frac{n}{2}\left(\frac{1}{1+\nu S} - \frac{1}{1+\nu D}\right)^2 \tag{353}$$

$$\leq C_1 \nu(\|e_0^n\|^2 + n), \tag{354}$$

where (353) uses the vanishing odd Gaussian moments and (354) uses $\|e^{n,\mathrm{c}}\|^2 \leq nD$ and $\nu \to 0$. Define the gain remainder by

$$\mathsf{R}_n^{\mathrm{g}} := \frac{\nu(\|e_0^n\|^2 - nS)}{2(1+\nu S)} + \mathsf{N}_n. \tag{355}$$

Since $e_0^n \sim \mathcal{N}(0_n, SI_n)$ under P, (354) and Chebyshev give, for every $\eta > 0$,

$$P\{\mathcal{S}_n, F_n = 1,\ |\mathsf{R}_n^{\mathrm{g}}| > \eta\sqrt{n}\} \longrightarrow 0. \tag{356}$$

Comparing the two Gaussian densities yields, on $\mathcal{S}_n \cap \{F_n = 1\}$,

$$\log \frac{\kappa(A^{n,\mathrm{ext}} \mid M_0, Y^n)}{p_P(A^{n,\mathrm{ext}} \mid Y^n)} = \frac{n}{2}\log\frac{1+\nu S}{1+\nu D} + \frac{\|A^{n,\mathrm{ext}} - \sqrt{\nu}\rho Y^n\|^2}{2(1+\nu S)} - \frac{\|A^{n,\mathrm{ext}} - \sqrt{\nu}\bar{X}^n\|^2}{2(1+\nu D)} \tag{357}$$

$$= n(R_{\mathrm{WZ}}(D) - \tilde{R}) + \frac{\nu}{2}\left(\frac{\|e_0^n\|^2 - nS}{1+\nu S} - \frac{\|e^{n,\mathrm{c}}\|^2 - nD}{1+\nu D}\right) + \mathsf{N}_n \tag{358}$$

$$\geq n(R_{\mathrm{WZ}}(D) - \tilde{R}) + \mathsf{R}_n^{\mathrm{g}}, \tag{359}$$

where (357) substitutes the densities just defined, (358) expands the squares and uses (113) and (352), and (359) uses $\|e^{n,\mathrm{c}}\|^2 \leq nD$ and (355). The actual gain factors as

$$\log \frac{p_P(A^{n,\mathrm{ext}} \mid M_0, Y^n)}{p_P(A^{n,\mathrm{ext}} \mid Y^n)} = \log \frac{p_P(A^{n,\mathrm{ext}} \mid M_0, Y^n)}{\kappa(A^{n,\mathrm{ext}} \mid M_0, Y^n)} + \log \frac{\kappa(A^{n,\mathrm{ext}} \mid M_0, Y^n)}{p_P(A^{n,\mathrm{ext}} \mid Y^n)}. \tag{360}$$

Therefore, for every $\eta > 0$,

$$P\left\{\mathcal{S}_n, F_n = 1,\ \log \frac{p_P(A^{n,\mathrm{ext}} \mid M_0, Y^n)}{p_P(A^{n,\mathrm{ext}} \mid Y^n)} < n(R_{\mathrm{WZ}}(D) - \tilde{R}) - \eta\sqrt{n}\right\} \longrightarrow 0, \tag{361}$$

where (361) uses (360), (359), (356), and (350) with $\gamma = \log n = o(\sqrt{n})$.

Define $\tilde{H}_0 := -\log P(M_0 \mid \tilde{Y}^n, \Pi)$, where the conditional mass is obtained by integrating $\varphi_{\tilde{S}I_n}(x^n - \tilde{\rho}\tilde{Y}^n)$ over the source cell of M_0, as in (116). Then

$$-\log P(M_0 \mid Y^n, \Pi) - \tilde{H}_0 = \log \frac{P(M_0 \mid Y^n, A^{n,\mathrm{ext}}, \Pi)}{P(M_0 \mid Y^n, \Pi)} \tag{362}$$

$$= \log \frac{p_P(A^{n,\mathrm{ext}} \mid M_0, Y^n)}{p_P(A^{n,\mathrm{ext}} \mid Y^n)}, \tag{363}$$

where (362) uses the invertible transformation (345) and independence of $V^{n,\mathrm{anc}}$, and (363) uses Bayes' rule and independence of Π from $(M_0, Y^n, A^{n,\mathrm{ext}})$. Since M_0 has at most $2M_n$ values, the finite-message likelihood bound in [7, Appendix C-C, Eqs. (300)–(302)] gives

$$P\{-\log P(M_0 \mid Y^n, \Pi) > \log(2M_n) + \gamma\} \leq e^{-\gamma}. \tag{364}$$

For every $\eta > 0$, it follows that

$$P\{\mathcal{S}_n, F_n = 1,\ \tilde{H}_0 > n\tilde{R} + \sqrt{n}(L + \eta)\} \longrightarrow 0, \tag{365}$$

where (365) combines (363), (364) with $\gamma = \log n$, (361), and the code budget (90). The term $n(R_{\mathrm{WZ}}(D) - \tilde{R})$ from (361) cancels the corresponding change in the first-order rate.

B. Sampled Description Cost

Define $\mathcal{B} := \sigma(M_0, \tilde{Y}^n, \Pi, V^{n,\mathrm{anc}})$. The reconstruction $\hat{X}^{n,\mathrm{enh}}$ is fixed given $\mathcal{B}$. Let $p_{\mathcal{B}}$ be the density $\varphi_{\tilde{S}I_n}(x^n - \tilde{\rho}\tilde{Y}^n)$ restricted and normalized on the source cell of M_0. Independence of $V^{n,\mathrm{anc}}$ makes $p_{\mathcal{B}}$ the conditional source density given $\mathcal{B}$. Let p^{pre} be the induced mass function of V^{pre}. The two conditional distributions agree under P and $\tilde{P}$ on cells with $F_n = 1$. Define the Gaussian reference $\kappa_{\mathcal{B}}(x^n) := \varphi_{\tilde{D}I_n}(x^n - \hat{X}^{n,\mathrm{enh}})$ and its induced mass function on the sampled description by

$$\kappa^{\mathrm{pre}}(v^{\mathrm{pre}} \mid \mathcal{B}) := \int_{\{x^n : V^{\mathrm{pre}}(x^n, \Pi) = v^{\mathrm{pre}}\}} \kappa_{\mathcal{B}}(x^n)\, dx^n. \tag{366}$$

Define the full and prefix log likelihood ratios by

$$\zeta := \log \frac{p_{\mathcal{B}}(X^n)}{\kappa_{\mathcal{B}}(X^n)} \tag{367}$$

and

$$\zeta_k := \log \frac{p^{\mathrm{pre}}(V^{\mathrm{pre}} \mid \mathcal{B})}{\kappa^{\mathrm{pre}}(V^{\mathrm{pre}} \mid \mathcal{B})}. \tag{368}$$

The difference $\zeta - \zeta_k$ is the log ratio of the corresponding conditional source densities given the prefix. Consequently,

$$\tilde{P}(\zeta_k < -\gamma) \le e^{-\gamma} \tag{369}$$

and

$$\tilde{P}(\zeta - \zeta_k < -\gamma) \le e^{-\gamma}, \tag{370}$$

where (369) applies (350) to ζ_k, and (370) applies the same bound to $\zeta - \zeta_k$.

Put $\tilde{e}^n := X^n - \hat{X}^{n,\mathrm{enh}}$, and define

$$E_n := \frac{\|\tilde{e}^n\|^2 - n\tilde{D}}{2\tilde{D}} \tag{371}$$

and

$$E_k := \frac{\sum_{j=1}^{k} \tilde{e}^2_{\Pi(j)} - k\tilde{D}}{2\tilde{D}}. \tag{372}$$

The flag, clipping, and the Gaussian maximum bound imply that, for some constant C_2 and all sufficiently large n,

$$\tilde{P}\{\|\tilde{e}^n\|_\infty > C_2\sqrt{\log n}\} \le 2n^{-3}. \tag{373}$$

Indeed, (346) bounds $\|\tilde{e}^n\|_\infty$ by $2h_n + D\sqrt{\nu}\|G^{n,\mathrm{ext}}\|_\infty$, and $\|G^{n,\mathrm{ext}}\|_\infty \le \sqrt{8\log n}$ fails with probability at most $2n^{-3}$.

For $j \in [k]$, let $\mathcal{I}_j$ be the realized quantization interval containing $X_{\Pi(j)}$. On $\{F_n = 1\}$, each interval $\mathcal{I}_j$ has width Δ_n by (100). Define the quantization remainder $\mathsf{R}_n^{\mathrm{q}}$ through the following exact calculation:

$$-\log \kappa^{\mathrm{pre}}(V^{\mathrm{pre}} \mid \mathcal{B}) = -\sum_{j=1}^{k} \log \int_{\mathcal{I}_j} \varphi_{\tilde{D}}(x - \hat{X}^{\mathrm{enh}}_{\Pi(j)})\, dx \tag{374}$$

$$= k \log \frac{1}{\Delta_n} - \sum_{j=1}^{k} \log \varphi_{\tilde{D}}(\tilde{e}_{\Pi(j)}) + \mathsf{R}_n^{\mathrm{q}} \tag{375}$$

$$= k \log \frac{1}{\Delta_n} + \frac{k}{2}\log(2\pi\tilde{D}) + \frac{1}{2\tilde{D}} \sum_{j=1}^{k} \tilde{e}^2_{\Pi(j)} + \mathsf{R}_n^{\mathrm{q}} \tag{376}$$

$$= b_n^{\mathrm{pre}} + E_k + \mathsf{R}_n^{\mathrm{q}}, \tag{377}$$

where (374) uses the product reference in (366), (375) defines $\mathsf{R}_n^{\mathrm{q}}$, (376) substitutes the scalar Gaussian density, and (377) uses (118) and (372). The log density varies by at most $\Delta_n(\|\tilde{e}^n\|_\infty + \Delta_n/2)/\tilde{D}$ within each interval. Thus (373), $k = \lceil n^{3/4} \rceil$, and $\Delta_n = n^{-2}$ give, for every $\eta > 0$,

$$\tilde{P}\{|\mathsf{R}_n^{\mathrm{q}}| > \eta\} \longrightarrow 0. \tag{378}$$

The estimate (378) approximates only the Gaussian reference mass. It does not require any regularity of the true posterior density within a quantization interval.

The error vector $\tilde{e}^n$ is independent of Π. For deterministic $z_1, \ldots, z_n$, write $\bar{z} := n^{-1}\sum_{i=1}^{n} z_i$. Uniform sampling without replacement gives

$$\mathrm{Var}_\Pi\left(\sum_{j=1}^{k} z_{\Pi(j)}\right) = \frac{k(n-k)}{n-1}\frac{1}{n}\sum_{i=1}^{n}(z_i - \bar{z})^2, \tag{379}$$

where (379) follows by summing the diagonal and off-diagonal covariances of the sampled coordinates. Therefore,

$$\mathrm{Var}_\Pi\left(E_k - \frac{k}{n}E_n \,\middle|\, \tilde{e}^n\right) = \frac{k(n-k)}{4\tilde{D}^2(n-1)}\frac{1}{n}\sum_{i=1}^{n}\left(\tilde{e}_i^2 - \frac{\|\tilde{e}^n\|^2}{n}\right)^2 \tag{380}$$

$$\le \frac{k}{4\tilde{D}^2}\|\tilde{e}^n\|_\infty^4, \tag{381}$$

where (380) uses (379) and (372), and (381) bounds the empirical variance by the second moment and uses $(n-k)/(n-1) \le 1$. The conditional mean is zero. For every $\eta > 0$, conditional Chebyshev and (373) yield

$$\tilde{P}\left\{\left|E_k - \frac{k}{n}E_n\right| > \eta\sqrt{n}\right\} \le 2n^{-3} + \frac{C_2^4 k(\log n)^2}{4\tilde{D}^2\eta^2 n} \tag{382}$$

$$\longrightarrow 0, \tag{383}$$

where (382) conditions on $\tilde{e}^n$ and bounds the exceptional maximum event, and (383) uses $k(\log n)^2 = o(n)$ and $\tilde{D} \to D > 0$. Combining (383) with (349), we obtain

$$P\{\mathcal{S}_n, F_n = 1,\ E_k > \eta\sqrt{n}\} \longrightarrow 0 \tag{384}$$

and

$$P\{\mathcal{S}_n, F_n = 1,\ E_k - E_n < -\eta\sqrt{n}\} \longrightarrow 0, \tag{385}$$

where (384) uses $k/n \le 1$, (385) uses $E_k - E_n = -(1-k/n)E_n + [E_k - (k/n)E_n]$, and both use (98) to pass from $\tilde{P}$ to P on the flag set.

C. Upper and Lower Likelihood Comparisons

We first prove (119). Factoring the conditional probability of $W = (M_0, V^{\mathrm{pre}})$ gives

$$\tilde{H} = \tilde{H}_0 - \log P(V^{\mathrm{pre}} \mid M_0, \tilde{Y}^n, \Pi) \tag{386}$$

$$= \tilde{H}_0 - \log \kappa^{\mathrm{pre}}(V^{\mathrm{pre}} \mid \mathcal{B}) - \zeta_k \tag{387}$$

$$= \tilde{H}_0 + b_n^{\mathrm{pre}} + E_k - \zeta_k + \mathsf{R}_n^{\mathrm{q}}, \tag{388}$$

where (386) uses $W = (M_0, V^{\mathrm{pre}})$, (387) uses independence of $V^{n,\mathrm{anc}}$ and (368), and (388) uses (377). For every $\eta > 0$,

$$P\{\mathcal{S}_n, F_n = 1,\ \tilde{H} > n\tilde{R} + b_n^{\mathrm{pre}} + \sqrt{n}(L+\eta)\} \longrightarrow 0, \tag{389}$$

where (389) applies the union bound to (388), using (365), (384), (369), and (378). This proves (119).

We next prove (120). Let $p_{X^n|\tilde{Y}^n}$ and $p_{X^n|M_0,\tilde{Y}^n,\Pi}$ denote, respectively, the original conditional Gaussian density and its normalized restriction to the source cell of M_0. Bayes' rule gives

$$\tilde{H}_0 = \log \frac{p_{X^n|M_0,\tilde{Y}^n,\Pi}(X^n \mid M_0, \tilde{Y}^n, \Pi)}{p_{X^n|\tilde{Y}^n}(X^n \mid \tilde{Y}^n)} \tag{390}$$

$$= \zeta + \log \frac{\kappa_{\mathcal{B}}(X^n)}{p_{X^n|\tilde{Y}^n}(X^n \mid \tilde{Y}^n)} \tag{391}$$

$$= \zeta + \frac{n}{2}\log\frac{\tilde{S}}{\tilde{D}} + \frac{\|X^n - \tilde{\rho}\tilde{Y}^n\|^2}{2\tilde{S}} - \frac{\|\tilde{e}^n\|^2}{2\tilde{D}} \tag{392}$$

$$= \zeta + n\tilde{R} + \frac{\|X^n - \tilde{\rho}\tilde{Y}^n\|^2 - n\tilde{S}}{2\tilde{S}} - E_n, \tag{393}$$

where (390) uses determinism of M_0 given X^n, (391) uses independence of $V^{n,\mathrm{anc}}$ and (367), (392) substitutes the two Gaussian densities, and (393) uses (111) and (371). Adding the prefix term yields

$$\tilde{H} = \tilde{H}_0 + b_n^{\mathrm{pre}} + E_k - \zeta_k + \mathsf{R}_n^{\mathrm{q}} \tag{394}$$

$$= n\tilde{R} + b_n^{\mathrm{pre}} + \frac{\|X^n - \tilde{\rho}\tilde{Y}^n\|^2 - n\tilde{S}}{2\tilde{S}} + (E_k - E_n) + (\zeta - \zeta_k) + \mathsf{R}_n^{\mathrm{q}}, \tag{395}$$

where (394) recalls (388), and (395) substitutes (393). For every $\eta > 0$,

$$P\left\{\mathcal{S}_n, F_n = 1,\ \tilde{H} < n\tilde{R} + b_n^{\mathrm{pre}} + \frac{\|X^n - \tilde{\rho}\tilde{Y}^n\|^2 - n\tilde{S}}{2\tilde{S}} - \eta\sqrt{n}\right\} \longrightarrow 0, \tag{396}$$

where (396) applies the union bound to (395), using (385), (370), and (378). This proves (120) and completes the proof of Lemma 4.

Appendix G
Proof of Lemma 5

Recall b^n and ℓ^n in (121)–(122), both determined by (X^n, Π). We first approximate the centered likelihood using inner products with the Gaussian noise and its centered squared norm. We then use the posterior relative entropy to prove the squared-norm and inner-product limits in (127)–(129). All derivatives use the conditional densities obtained by integrating the Gaussian channel density over source cells. We consider only source cells having positive probability.

A. Approximation of the Centered Likelihood

Recall the source cell $\mathcal{C}_{w,\pi}$ defined above (101). Write $p_{X^n} = \varphi_{pI_n}$ and $p_{\tilde{Y}^n} = \varphi_{(p+\tilde{q})I_n}$ for the original source and enhanced-observation densities. On a retained cell of positive probability, Bayes' rule gives

$$P(W = w \mid \tilde{Y}^n = y^n, \Pi = \pi) = \frac{\int_{\mathcal{C}_{w,\pi}} p_{X^n}(x^n)\varphi_{\tilde{q}I_n}(y^n - x^n)\,dx^n}{p_{\tilde{Y}^n}(y^n)}, \tag{397}$$

where the numerator in (397) integrates the original Gaussian channel density over $\mathcal{C}_{w,\pi}$. Define $m^n(w, y^n, \pi) := \mathsf{E}_{\tilde{P}}[X^n \mid W = w, \tilde{Y}^n = y^n, \Pi = \pi]$ and let $V(w, y^n, \pi)$ be the covariance under the same conditional distribution. At the realized values $(W, \tilde{Y}^n, \Pi)$, write m^n and V for the mean and covariance. Since F_n is a coordinate of W, the posterior on a cell with $F_n = 1$ agrees under P and $\tilde{P}$. Differentiating the conditional message probability gives

$$\nabla_{y^n}\tilde{H} = \frac{\tilde{\rho}y^n - m^n}{\tilde{q}}, \tag{398}$$

where (398) follows from [15, Proposition 11, Eq. (99)], applied to $\mathcal{C}_{w,\pi}$, and $\mathsf{E}_P[X^n \mid \tilde{Y}^n = y^n] = \tilde{\rho}y^n$. The conditional-mean derivative identity gives

$$\nabla_{y^n} m^n = \frac{V}{\tilde{q}}, \tag{399}$$

where (399) follows from [15, Proposition 1, Eq. (17)] under the conditional input distribution on the cell.

Define $v^n(y^n) := m^n(W, y^n, \Pi) - \tilde{D}y^n/\tilde{q}$, and let $r_m^n := v^n(\tilde{Y}^n) - \mathsf{E}_{\tilde{P}}[v^n(\tilde{Y}^n) \mid X^n, \Pi]$. From (399),

$$\nabla_{y^n} v^n = \frac{V - \tilde{D}I_n}{\tilde{q}}. \tag{400}$$

Define the centered likelihood residual by

$$R_n := \tilde{H} - \mathsf{E}_{\tilde{P}}[\tilde{H} \mid X^n, \Pi] - (\ell^n)^\top \tilde{Z}^n - \frac{\tilde{\rho} - \tilde{D}/\tilde{q}}{2\tilde{q}}(\|\tilde{Z}^n\|^2 - n\tilde{q}). \tag{401}$$

Conditioning on (X^n, Π) fixes W, so $\mathsf{E}_{\tilde{P}}[R_n \mid X^n, \Pi] = 0$ by centering and the Gaussian noise moments. Differentiating with (X^n, Π) fixed gives

$$\nabla_{y^n} R_n = \frac{\tilde{\rho}y^n - m^n}{\tilde{q}} - \ell^n - \frac{\tilde{\rho} - \tilde{D}/\tilde{q}}{\tilde{q}}(y^n - X^n) \tag{402}$$

$$= -\frac{m^n - \tilde{D}y^n/\tilde{q} - (X^n - b^n - \tilde{D}X^n/\tilde{q})}{\tilde{q}} \tag{403}$$

$$= -\frac{v^n(\tilde{Y}^n) - \mathsf{E}_{\tilde{P}}[v^n(\tilde{Y}^n) \mid X^n, \Pi]}{\tilde{q}}, \tag{404}$$

where (402) uses (398) and (401), (403) substitutes (122), and (404) uses (121) and $\mathsf{E}_{\tilde{P}}[\tilde{Y}^n \mid X^n, \Pi] = X^n$. The gradients are evaluated at $y^n = \tilde{Y}^n$ in the last line.

The Gaussian Poincaré inequality [16, Sec. 3.1, Eq. (3.2), p. 49], first applied to R_n and then componentwise to v^n under $\tilde{Y}^n \mid X^n \sim \mathcal{N}(X^n, \tilde{q}I_n)$, yields

$$\mathsf{E}_{\tilde{P}}[R_n^2 \mid X^n, \Pi] \le \tilde{q}\mathsf{E}_{\tilde{P}}[\|\nabla_{y^n} R_n\|^2 \mid X^n, \Pi] \tag{405}$$

$$= \tilde{q}^{-1}\mathsf{E}_{\tilde{P}}[\|v^n - \mathsf{E}_{\tilde{P}}[v^n \mid X^n, \Pi]\|^2 \mid X^n, \Pi] \tag{406}$$

$$\le \mathsf{E}_{\tilde{P}}[\|\nabla_{y^n} v^n\|_{\mathrm{F}}^2 \mid X^n, \Pi] \tag{407}$$

$$= \tilde{q}^{-2}\mathsf{E}_{\tilde{P}}[\|V - \tilde{D}I_n\|_{\mathrm{F}}^2 \mid X^n, \Pi], \tag{408}$$

where (405) uses conditional centering and Gaussian Poincaré, (406) uses (404), (407) applies the same inequality to the coordinates of v^n, and (408) uses (400). The retained source support is bounded, so the posterior means and their Gaussian-smoothed derivatives satisfy the required integrability conditions. Averaging gives

$$\mathsf{E}_{\tilde{P}}[R_n^2] = \mathsf{E}_{\tilde{P}}[\mathsf{E}_{\tilde{P}}[R_n^2 \mid X^n, \Pi]] \tag{409}$$

$$\le \tilde{q}^{-2}\mathsf{E}_{\tilde{P}}[\|V - \tilde{D}I_n\|_{\mathrm{F}}^2] \tag{410}$$

$$= o(n), \tag{411}$$

where (409) is the tower property, (410) uses (408), and (411) uses (104) and $\tilde{q} \to q > 0$. Chebyshev's inequality proves (126).

The same componentwise estimate gives

$$\mathsf{E}_{\tilde{P}}[\|r_m^n\|^2] = \mathsf{E}_{\tilde{P}}[\|v^n(\tilde{Y}^n) - \mathsf{E}_{\tilde{P}}[v^n(\tilde{Y}^n) \mid X^n, \Pi]\|^2] \tag{412}$$
$$\leq \tilde{q}\mathsf{E}_{\tilde{P}}[\|\nabla_{y^n} v^n\|_{\mathrm{F}}^2] \tag{413}$$
$$= \tilde{q}^{-1}\mathsf{E}_{\tilde{P}}[\|V - \tilde{D}I_n\|_{\mathrm{F}}^2] \tag{414}$$
$$= o(n), \tag{415}$$

where (412) is the definition of r_m^n, (413) is conditional Gaussian Poincaré followed by averaging, (414) uses (400), and (415) uses (104).

B. Limits of Squared Norms and Inner Products

Let $e^n := X^n - m^n$ be the posterior estimation error. Its decomposition is

$$e^n = X^n - \mathsf{E}_{\tilde{P}}[m^n \mid X^n, \Pi] - \big(m^n - \mathsf{E}_{\tilde{P}}[m^n \mid X^n, \Pi]\big) \tag{416}$$
$$= b^n - \frac{\tilde{D}}{\tilde{q}}\tilde{Z}^n - r_m^n, \tag{417}$$

where (416) adds and subtracts the conditional mean, and (417) uses b^n in (121) and the definitions of v^n and r_m^n above (400). Since $b^n = \mathsf{E}_{\tilde{P}}[e^n \mid X^n, \Pi]$,

$$\mathsf{E}_{\tilde{P}}[\|b^n\|^2] = \mathsf{E}_{\tilde{P}}[\|\mathsf{E}_{\tilde{P}}[e^n \mid X^n, \Pi]\|^2] \tag{418}$$
$$\leq \mathsf{E}_{\tilde{P}}[\|e^n\|^2] \tag{419}$$
$$= \mathsf{E}_{\tilde{P}}[\|X^n\|^2] - \mathsf{E}_{\tilde{P}}[\|m^n\|^2] \tag{420}$$
$$\leq \mathsf{E}_{\tilde{P}}[\|X^n\|^2] \tag{421}$$
$$\leq 2pn, \tag{422}$$

where (418) uses the conditional-mean representation of b^n, (419) is Jensen's inequality, (420) is conditional-mean orthogonality, (421) drops a nonnegative term, and (422) uses the source flag. Pointwise on the conditioning support,

$$\|m^n\|^2 \leq \mathsf{E}_{\tilde{P}}[\|X^n\|^2 \mid W, \tilde{Y}^n, \Pi] \tag{423}$$
$$\leq 2pn, \tag{424}$$

where (423) is conditional Jensen and (424) again uses the flag.

To determine the squared norm of e^n and its inner product with m^n, define a reference distribution $\bar{P}$ with the same $(W, \tilde{Y}^n, \Pi)$ marginal as $\tilde{P}$ and conditional source distribution $\mathcal{N}(m^n, \tilde{D}I_n)$. Then

$$\mathrm{KL}(\tilde{P}\|\bar{P}) = \mathsf{E}_{\tilde{P}}[\mathrm{KL}(\tilde{P}_{X^n|W,\tilde{Y}^n,\Pi}\|\mathcal{N}(m^n, \tilde{D}I_n))] \tag{425}$$
$$= K(t_0 + \nu_n) \tag{426}$$
$$= o(n), \tag{427}$$

where (425) is the relative entropy chain rule with the common marginal, (426) uses the sufficient enhanced observation and (102), and (427) uses (103). For any event $\mathcal{E}$ with $\bar{P}(\mathcal{E}) \in (0, 1)$, binary data processing gives

$$\mathrm{KL}(\tilde{P}\|\bar{P}) \geq \tilde{P}(\mathcal{E}) \log \frac{1}{\bar{P}(\mathcal{E})} - \log 2, \tag{428}$$

where (428) bounds the binary entropy by $\log 2$ and drops the other nonnegative cross-entropy term. In particular, if $\bar{P}(\mathcal{E})$ decreases exponentially in n, the $o(n)$ relative entropy in (427) implies $\tilde{P}(\mathcal{E}) \to 0$.

Fix $\eta > 0$. Under $\bar{P}$, e^n is conditionally Gaussian with covariance $\tilde{D}I_n$. Since $\tilde{D} \to D > 0$, the chi-square tail bound supplies a constant $c_\eta > 0$ such that

$$\bar{P}\left(\left|\frac{\|e^n\|^2}{n} - \tilde{D}\right| > \eta\right) \leq 2e^{-c_\eta n} \tag{429}$$

for all sufficiently large n. Combining (428) and (429) gives

$$\tilde{P}\left(\left|\frac{\|e^n\|^2}{n} - \tilde{D}\right| > \eta\right) \leq \frac{K(t_0 + \nu_n) + \log 2}{c_\eta n - \log 2} \tag{430}$$
$$= o(1), \tag{431}$$

where (430) uses (428) and (429), and (431) uses (103). Consequently,

$$\frac{\|e^n\|^2}{n} - \tilde{D} \longrightarrow 0 \quad \text{in } \tilde{P}\text{-probability.} \tag{432}$$

The pointwise bound (424) holds under both $\tilde{P}$ and $\bar{P}$ because the two distributions have the same $(W, \tilde{Y}^n, \Pi)$ marginal. Therefore,

$$\bar{P}(|(m^n)^\top e^n| > \eta n) \le 2\mathsf{E}_{\bar{P}}\left[\exp\left\{-\frac{\eta^2 n^2}{2\tilde{D}\|m^n\|^2}\right\}\right] \tag{433}$$

$$\le 2\exp\{-\eta^2 n/(4p\tilde{D})\}, \tag{434}$$

where (433) conditions on $(W, \tilde{Y}^n, \Pi)$ and uses the Gaussian tail bound, with zero contribution when $m^n = 0$, and (434) uses (424). Applying (428) again gives $(m^n)^\top e^n/n \to 0$ in $\tilde{P}$-probability. Consequently, also in $\tilde{P}$-probability,

$$\frac{(X^n)^\top e^n}{n} - \tilde{D} = \frac{(m^n)^\top e^n}{n} + \left(\frac{\|e^n\|^2}{n} - \tilde{D}\right) \tag{435}$$

$$\longrightarrow 0, \tag{436}$$

where (435) uses $e^n = X^n - m^n$, and (436) uses (432) and the transferred orthogonality estimate.

Conditioned on (X^n, Π), the projections $(X^n)^\top \tilde{Z}^n/n$ and $(b^n)^\top \tilde{Z}^n/n$ are centered Gaussian. Their averaged variances are at most $2p\tilde{q}/n$ by (93) and (422), so both converge to zero in $\tilde{P}$-probability. Also, $\|\tilde{Z}^n\|^2/n - \tilde{q} \to 0$ by the Gaussian variance bound, and $\|r_m^n\|/\sqrt{n} \to 0$ by (415). Therefore, in $\tilde{P}$-probability,

$$\frac{(X^n)^\top b^n}{n} - \tilde{D} = \left(\frac{(X^n)^\top e^n}{n} - \tilde{D}\right) + \frac{\tilde{D}}{\tilde{q}}\frac{(X^n)^\top \tilde{Z}^n}{n} + \frac{(X^n)^\top r_m^n}{n} \tag{437}$$

$$\longrightarrow 0, \tag{438}$$

where (437) rearranges (417), and (438) uses (436), the projection limit, and Cauchy–Schwarz with $\|X^n\|^2 \le 2pn$. Similarly,

$$\frac{\|b^n\|^2}{n} + \frac{\tilde{D}^2}{\tilde{q}} - \frac{\|e^n\|^2}{n} = \frac{\|b^n\|^2 - \|b^n - (\tilde{D}/\tilde{q})\tilde{Z}^n - r_m^n\|^2}{n} + \frac{\tilde{D}^2}{\tilde{q}} \tag{439}$$

$$= -\left(\frac{\tilde{D}}{\tilde{q}}\right)^2\left(\frac{\|\tilde{Z}^n\|^2}{n} - \tilde{q}\right) + 2\frac{\tilde{D}}{\tilde{q}}\frac{(b^n)^\top \tilde{Z}^n}{n} + \frac{2(r_m^n)^\top(b^n - (\tilde{D}/\tilde{q})\tilde{Z}^n) - \|r_m^n\|^2}{n} \tag{440}$$

$$\longrightarrow 0, \tag{441}$$

where (439) uses (417), (440) expands the square, and (441) follows in $\tilde{P}$-probability from the noise-energy and projection limits, Cauchy–Schwarz, (415), and (422). Thus (438) proves (128), while (432) and (441) prove (129).

Finally, for every $\eta > 0$,

$$\tilde{P}\left(\left|\frac{\|X^n\|^2}{n} - p\right| > \eta\right) \le \frac{P(|\|X^n\|^2/n - p| > \eta)}{\beta_n} \tag{442}$$

$$\longrightarrow 0, \tag{443}$$

where (442) uses the definition of $\tilde{P}$, and (443) uses the Gaussian law of large numbers and β_n bounded away from zero. This proves (127) and completes the proof of Lemma 5.

REFERENCES

[1] C. E. Shannon, "Coding theorems for a discrete source with a fidelity criterion," in *IRE International Convention Record*, vol. 7, 1959, pp. 142–163, pt. 4.

[2] A. Ingber and Y. Kochman, "The dispersion of lossy source coding," in *Proc. Data Compression Conf. (DCC)*, 2011, pp. 53–62.

[3] V. Kostina and S. Verdú, "Fixed-length lossy compression in the finite blocklength regime," *IEEE Trans. Inf. Theory*, vol. 58, no. 6, pp. 3309–3338, Jun. 2012.

[4] A. D. Wyner and J. Ziv, "The rate-distortion function for source coding with side information at the decoder," *IEEE Trans. Inf. Theory*, vol. 22, no. 1, pp. 1–10, Jan. 1976.

[5] A. D. Wyner, "The rate-distortion function for source coding with side information at the decoder—II: General sources," *Information and Control*, vol. 38, no. 1, pp. 60–80, Jul. 1978.

[6] S.-Q. Le, V. Y. F. Tan, and M. Motani, "Second-order coding rates for conditional rate-distortion," 2014, arXiv:1410.2687v1.

[7] L. Zhou, "Exact second-order asymptotics for the discrete Wyner–Ziv problem," 2026, arXiv:2610.11998v1.

[8] S. Watanabe, S. Kuzuoka, and V. Y. F. Tan, "Nonasymptotic and second-order achievability bounds for coding with side-information," *IEEE Trans. Inf. Theory*, vol. 61, no. 4, pp. 1574–1605, Apr. 2015.

[9] C. T. Li and V. Anantharam, "A unified framework for one-shot achievability via the Poisson matching lemma," *IEEE Trans. Inf. Theory*, vol. 67, no. 5, pp. 2624–2651, May 2021.

[10] X. Li and C. T. Li, "New second-order achievability bounds for coding with side information via type deviation convergence," in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)*, Guangzhou, China, Jul. 2026.

[11] D. Guo, S. Shamai (Shitz), and S. Verdú, "Mutual information and minimum mean-square error in Gaussian channels," *IEEE Trans. Inf. Theory*, vol. 51, no. 4, pp. 1261–1282, Apr. 2005.

[12] M. Payaró and D. P. Palomar, "Hessian and concavity of mutual information, differential entropy, and entropy power in linear vector Gaussian channels," *IEEE Trans. Inf. Theory*, vol. 55, no. 8, pp. 3613–3628, Aug. 2009.

[13] W.-K. Chen, N. Dafnis, and G. Paouris, "Improved Hölder and reverse Hölder inequalities for Gaussian random vectors," *Adv. Math.*, vol. 280, pp. 643–689, Aug. 2015.

[14] W. Hoeffding, "Probability inequalities for sums of bounded random variables," *J. Amer. Statist. Assoc.*, vol. 58, no. 301, pp. 13–30, Mar. 1963.
[15] A. Dytso, H. V. Poor, and S. Shamai (Shitz), "Conditional mean estimation in Gaussian noise: A meta derivative identity with applications," *IEEE Trans. Inf. Theory*, vol. 69, no. 3, pp. 1883–1898, Mar. 2023.
[16] M. Ledoux, *The Concentration of Measure Phenomenon*, ser. Mathematical Surveys and Monographs. Providence, RI, USA: American Mathematical Society, 2001, vol. 89.